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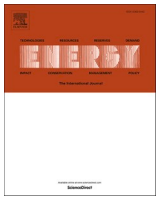

Study on the load-following ability of HeXe cooled SMR with close Brayton cycle for renewable energy integration

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ABSTRACT

Microgrids are vital for enhancing energy resilience and reliability, and the aim of carbon neutrality accelerates the integration of renewable energy. However, the variability of renewable energy and fluctuating demand present challenges to supply-load balance for renewable microgrids. This paper addresses the supply-load balance issues by integrating small modular reactors (SMRs) into renewable microgrids, developing load-following strategies, and evaluating transient performance under fluctuations. It presents a one-dimensional dynamic model of the Helium-Xenon cooled SMR with closed Brayton cycle (CBC) and transient simulations are conducted for three different control methods, to investigate the mechanisms and evaluate their characteristics and deficiencies. Combined regulation strategies are proposed, and their effectiveness is proven. From the evaluation, inventory control performs better initially. The turbine bypass valve control is also a favorable choice, especially under strict safety requirements. Using load regulation combinations, the system demonstrates a quick and fluctuation-free load-following in 20 %/100s ramp change and the 20 % step change within a 10-s offset for 100%–0% full range, able to manage the variability and demand fluctuations of microgrids. The outcomes of this study confirm the load-following capability of the HeXe cooled SMRs and highlight the feasibility and effectiveness of integrating nuclear energy into renewable microgrids.

1. Introduction

Microgrids are increasingly recognized for their importance in enhancing energy resilience and reliability, especially in isolated locations and areas recovering from natural disasters [1]. Meanwhile, as the world strives to achieve carbon neutrality and many countries announcing pledges to achieve net zero emissions over the coming decades [2], the development of low-carbon energy systems has become a critical priority, particularly in the context of the global "dual carbon" [3] (carbon peaking and carbon neutrality) strategy. Therefore, the future microgrids will use more energy from renewable sources, which will extend even more electricity to rural and remote areas that lack access to modern energy services [4].

The inherent variability and intermittency of renewable energy sources, such as solar and wind power, present significant challenges to maintaining a stable and continuous power supply: they are easily impacted by abnormal weather conditions [5] and mid-term fluctuations will inevitably increase significantly [6]; Power system operators are inclined to regard them as "disturbing elements" or "uncertain elements" for the operation of the power grid [7]. Higher penetrations lead to higher variability and uncertainty, affecting the microgrid's reliability and flexibility, and challenging supply-load balance.

To address these challenges, it is necessary to introduce other energy sources that can provide stable and reliable power output. Small modular reactors (SMRs) have emerged as a promising solution [8], which can provide low-carbon and long-term power with their long refueling cycles [9], while other common power sources may have some issues about emissions of CO_2, NOx, limitations to certain sites or difficulties in fuel supply. Therefore, whether the integration of SMR can benefit microgrids by flexibly adjusting their power output in response to changes in both renewable energy generation and consumer demand is urgently needed to be studied.

1.1. Challenges of supply-load balance for renewable microgrids

To maintain the supply-load balance in renewable microgrids, numerous studies have been conducted using various methodologies, listed in Table 1. However, there remain significant knowledge gaps that require further investigation. Existing research predominantly focuses

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Nomenclature

Symbols

A	Area[m^2]
C_i	Density distribution of delayed neutron pioneers ,$i = 1,2,\bullet\bullet\bullet,6$
C_p	Specific heat capacity [J/kg/K]
h	Heat transfer coefficient
$\dot{m}$	Mass flow rate
M	Mass
n	Neutron density
N	Rotational speed of turbine rotating machinery
p	Pressure
P	Power
Q	Thermal power of the reactor
R_0	Gas constant at standard state
V	Volume

Greek characters

β_i	Delayed neutron share ,$i = 1,2,\bullet\bullet\bullet,6$
β	Total share of delayed neutrons
λ_i	Decay constant ,$i = 1,2,\bullet\bullet\bullet,6$
Λ	Neutron time per generation
γ	Specific heat ratio
ρ	Reactivity
r_{tip}	Tip Radius of the turbomachinery

Subscripts

alt	Alternator
avr	Average
com	Compressor
core	Reactor core
cycle	Close Brayton cycle
in	Inlet
out	Outlet
rcp	Recuperator
s	Surge point
sys	System
tank	Storage tank
tur	Turbine

Abbreviation

CBC	Closed Brayton cycle
DRP	Demand response program
EMS	Energy Management System
ESS	Energy storage system
PR	Pressure ratio
RCBC	Recompression closed Brayton cycle
sCO_2	Supercritical carbon dioxide
SMR	Small modular reactor
TR	Temperature ratio
V1	The compressor bypass valve
V2	The turbine bypass valve
V3	The compressor return valve

on the reliability and flexibility of the system at the macro level, typically addressing steady-state characteristics. Detailed modeling for transient simulations under rapid changes in output power is rarely investigated, as well as specific load-following operating or control strategies for the hybrid power system in renewable microgrids. Additionally, supply-load balance has often been simplified to the question of whether electricity supply exceeds demand. However, several related issues need attention, such as whether the output power can respond to demand fluctuations without delay, and the time required for the system to stabilize following a drastic adjustment.

Some studies have addressed load-following strategies. To ensure that the energy flow through the microgrid is stabilized even under large-scale photovoltaic power fluctuations, Shin'ya [10] established an independent microgrid of an integrated coal gasification combined cycle (IGCC) and renewable energy. Then a PID controller is installed in the IGCC system and its load-following operation is studied. ZHICHAO ZHOU [11] proposed an independent microgrid, consisting of biomass power plants and wind turbine generators. The load following of biomass power plants and other methods are introduced to decrease the peak-load shifting demands and realize emergency power control for large disturbance scenarios.

1.2. Load-following strategies for SMRs in the renewable microgrids

The integration of SMRs and renewable energy offers a promising solution for enhancing energy reliability and high efficiency. A proposed hybrid nuclear-solar power system significantly mitigates the fluctuation of heat generation and reduces the discordance between the planned heat generation and the thermal load require d by the demand [25]. Temiz [26] developed a solar and reactor-based integrated energy system with the efficiency of 63.54 %, and enabled the community to become 100 % self-sufficient. Among the existing nuclear technologies, the Helium-Xenon (HeXe) cooled SMR with closed Brayton cycle (CBC) stands out as a potential one to be integrated into microgrids, due to its excellent thermodynamic properties and chemical stability [27]. Using an inert gas mixture as its coolant, HeXe cooled SMRs minimize corrosion issues and reduce system complexity by reducing the number of required compressors and recuperators [28].

As mentioned above, for the seamless integration of nuclear energy into renewable microgrids, investigation is needed on whether SMRs possess excellent load-following capabilities, which is mainly impacted by control strategies. Control strategies for gas-cooled SMR with CBC systems can be broadly categorized into valve control, heat source control, inventory control, and rotating speed control of turbomachinery. Relevant studies are summarized in Table 2.

Tri Q. Trinh [29] proposed various control methods for the supercritical CO_2 (sCO_2) recompression closed Brayton cycle (RCBC), mitigating a 10 % LOEL (Loss of External Load) accident. Moisseytsev [30] studied the variations in key parameters of the sCO_2 RCBC during the transient process involving a 10 % step and a 100 % linear decrease in the output power. Ming [31] proposed bypass, inventory, and combined control strategies in a sCO_2 CBC system and evaluated performance of different control strategies. Wang [32] used different control strategies to achieve the transient ramp change of load from 100 % to 0 % and compared the differences in efficiency and system performance under these strategies. Bian [33]investigated the performance of four different sCO_2 Brayton cycle layouts under various control methods to achieve system output power decreasing with a 10 %/min ramp. Liese [34] used inventory control for a 10 MWe sCO_2 RCBC to achieve a load decrease from 10 MWe to 4 MWe at an aggressive setpoint ramp rate of 0.75 MWe/min. Qiu [35] derived a dynamic model of the 5 MWe Helium RCBC and proposed control methods to achieve load-following with a 33 % step drop from the rated condition. El-Genk [36] established a HeXe space reactor CBC and applied PID control to the voltage of the starter motor, realizing the starting transient simulation. To study the control characteristics of the space Brayton cycle under part-load conditions, Li [37] et al. established a space HeXe CBC and explored the open-loop response for the bypass valve opening and changes in rotational speed.

As shown in Table 2, there are few transient studies for control

Table 1
Summary of reviewed papers about supply-load balance for renewable microgrids.

Ref.	Indictors and influence	Methods			
		DRP	ESS	EMS	Other
[12]	Reduce downtime and self-healing costs	●	●		
[13]	Increase a notable portion of the flexible capacities		●		Participation of Energy Communities and Internet Data Centers
[14]	Increase the stability of energy supply; reduce Forced Load Shedding (FLS)		●		Smart Grid Operator conducts; self-healing
[15]	Reduce Local Congestion Management Market (LCMM) costs	●			
[16]	Reduce the locational marginal price (LMP)	●	●		
[17]	Reduce flexibility locational marginal price	●		●	
[18]	The stability of power supply		●		Load shedding
[19]	Reduce load curtailment costs; balances of the power demand				Dispatch model
[20]	Reduce power loss and the voltage profile	●			
[21]	Zero load curtailment			●	EMS; Shifting the peak load demand to off-peak hours
[22]	Reduce the amount of energy not supplied (ENS)		●		Fossil fuel unit used as a support for reliable supply
[23]	Balance the power level in generation and request			●	Forecast and dispatch
[24]	Smooth load profile	●			Coordinated scheduling
[10]	Load fluctuation; load-following ability				Put load-following operation of IGCC into microgrid
[11]	Decrease the peak-load shifting demands				Add biomass power plants with load-following ability

Note: DRP: Demand response programs; ESS: Energy storage systems; EMS: Energy Management System.

strategies that cover a wide range of load demands or focus on transient responses of key parameters under different methods at full range. Secondly, most of the reviewed studies use the control method separately, less attention is paid to the combined method, which may improve the system's performance and safety, under part-load conditions. Additionally, while there are some studies investigating the load-following control of the integrated Pressurized Water Reactor (iPWR) type SMRs cooperating with other components of the microgrids to meet load demands [38], the interconnection technology of gas-cooled reactors with CBC and renewable energy microgrids has hardly been reported.

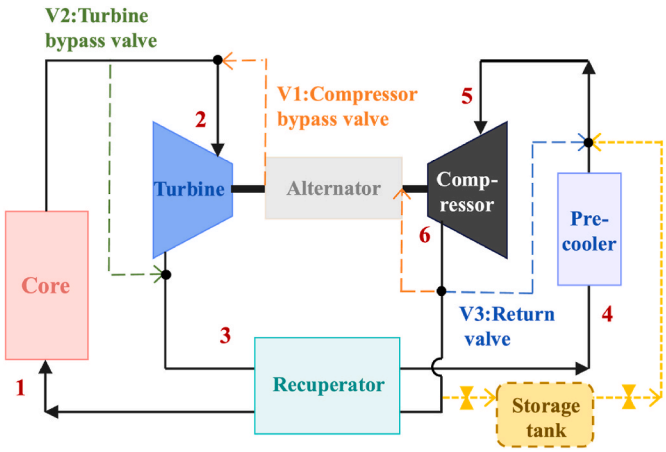


Fig. 1. The schematic layout of simple close Brayton cycle.

1.3. Contributions of this paper

This paper addresses the supply-load balance issues for renewable microgrids, especially load-following strategies, and transient performance in response to fluctuations in both renewable energy generation and demand. It presents a dynamic model of the HeXe cooled SMR with CBC and proposes load regulation combinations of different control

Table 3
The design steady-state condition of the system.

	Temperature[K]	Pressure [MPa]
Compressor inlet	330.00	1.00
Recuperator cold-side inlet	475.20	2.17
Reactor core inlet	932.00	2.10
Turbine inlet	1200.00	2.10
Recuperator hot-side inlet	979.40	1.09
Precooler inlet	522.30	1.00
Mass flow rate [kg/s]	10	
mass[kg]	8.152	
Thermal power[MW]	3.88	
Load power[MW]	1.15	
System efficiency	29.77 %	

Table 2
Summary of reviewed papers about CBC control strategies.

Ref	Object		Control method					Simulation		Performance	
	Fluid	System	V	H	T	R	Other	Open	Closed	Load-following	Other
[29]	sCO2	RCBC	●	●	●	●					mitigate a 10 % LOEL accident
[30]	sCO2	RCBC	●		●	●		●	●	**Ramp:** 100-0% at 10 %/min	
[31]	sCO2	CBC	●		●		V + T		●	**Step:** 20 % match in 450s; 100 % match in 3200s	
[32]	sCO2	RCBC	●		●		V + T	●	●	**Ramp:** 100-0% at 20 % in 50–100s	
[33]	sCO2	RCBC	●					●	●	**Step:** 10 % without apparent delay	
[34]	sCO2	RCBC			●				●	**Ramp:** 100%–40 % at 7.5 %/min	
[39]	sCO2	CBC			●			●	●	**Step:** 20 % match in 2000s, with ±9.45 % oscillation	
[35]	He	RCBC	●	●	●	●				**Step:** 100%–67 %	
[36]	HeXe	CBC				●					Start-up simulation
[37]	HeXe	CBC	●			●		●			mitigate load reduction and rejection
[40]	air	CBC				●		●	●	**Step:** 100%–0% match in 1500s	
[41]	air	OBC		●		●				**Step:** 10 % match in about 530s; range: 100%–30 %	

Note: V: Valve control; H: Heat source control; T: inventory control; R: rotating speed control of turbomachinery; Open: Open-loop; Closed: Closed-loop; OBC: Open Brayton cycle.

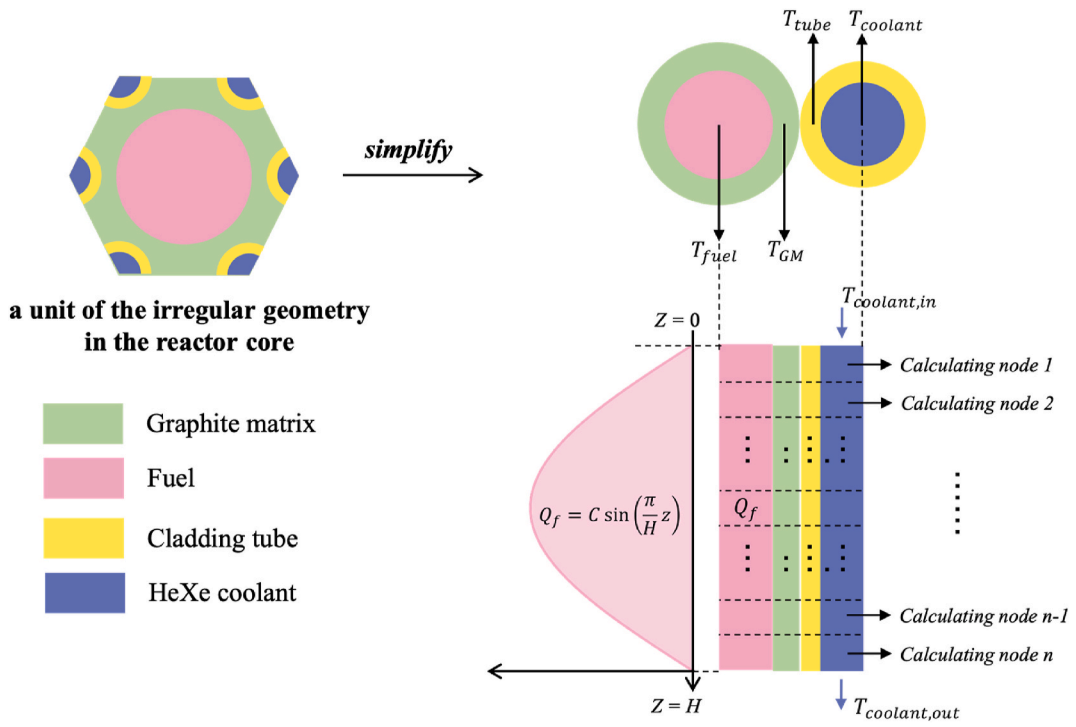


Fig. 2. One-dimensional calculating model of the reactor core.

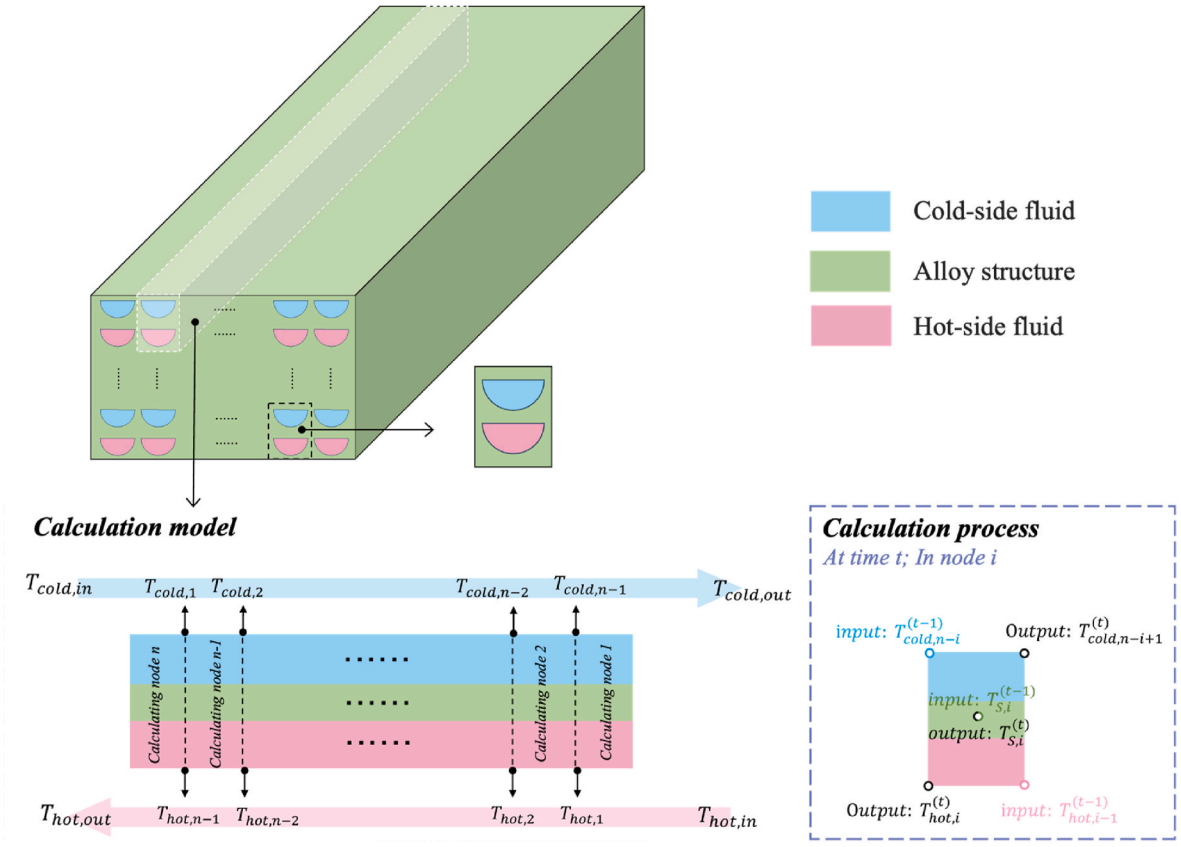


Fig. 3. Calculation schema of the heat exchanger.

methods for 100%–0% full range aiming to evaluate the system's ability to swiftly and stably modulate output power to meet the load-following requirements of microgrids. The main contributions of this work are summarized as follows.

Table 4
Verification of operating parameters of the system.

Variable name	Design value	Calculated value	Relative error
Compressor inlet temperature[K]	330.00	330.00	0.00 %
Recuperator cold-side inlet temperature[K]	475.20	474.00	0.25 %
Reactor core inlet temperature[K]	932.00	928.65	0.36 %
Turbine inlet temperature[K]	1200.00	1200.39	−0.03 %
Recuperator hot-side inlet temperature[K]	979.40	975.49	0.40 %
Precooler inlet temperature[K]	522.30	520.85	0.28 %
Compressor inlet pressure [MPa]	1.00	1.00	0.00 %
Recuperator cold-side inlet pressure [MPa]	2.17	2.17	0.00 %
Reactor core inlet pressure [MPa]	2.10	2.10	0.00 %
Turbine inlet pressure [MPa]	2.10	2.10	0.00 %
Recuperator hot-side inlet pressure [MPa]	1.09	1.09	0.00 %
Precooler inlet pressure [MPa]	1.00	1.00	0.00 %
Mass flow rate [kg/s]	10.00	9.99	0.10 %

- Modeling of different components and their coupling treatments for the proposed system is realized in the Python platform enabling accurate description and rigorous solution.
- Examination of the mechanisms of different control methods and their effects on critical parameters of the HeXe CBC system through open-loop simulations. Then, investigate the characteristics and limitations of control methods with safety and operational constraints through closed-loop simulations.
- Apply these load-following strategies in ramp or step curves to demonstrate its ability for rapid and stable output power adjustment, for evaluating the feasibility and effectiveness of integrating nuclear energy into renewable microgrids.

In this study, a one-dimensional dynamic model of a HeXe cooled reactor with CBC is proposed in Section 2. The correctness of the program is verified in Section 3. In Section 4, open-loop simulations are performed to analyze the mechanisms of different control strategies. The performance of different strategies is compared through closed-loop simulations. After that, regulation combinations for 100%–0% full range are proposed and the load-following ability to manage the variability and demand fluctuations is evaluated.

2. Modeling

In this section, the overall structure of the system and detailed modeling methodology for each component are introduced.

2.1. System description

In this study, a closed Brayton cycle dynamic model of 1 MW is established. The detailed design is described in the previous work [42], and the geometric parameters of the components used in the calculation are derived by using the optimization method in the literature [43].

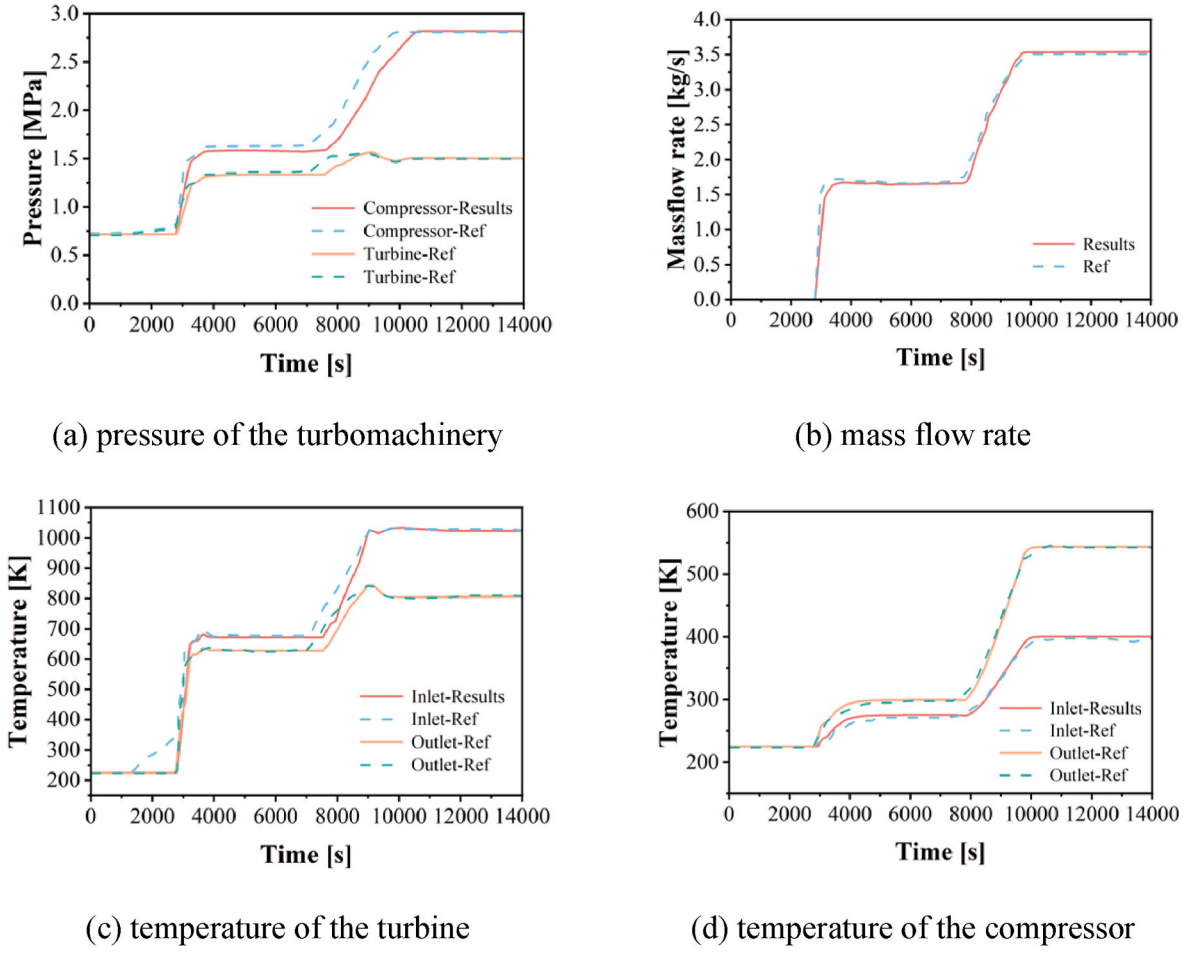


Fig. 4. The simulation and reference results.

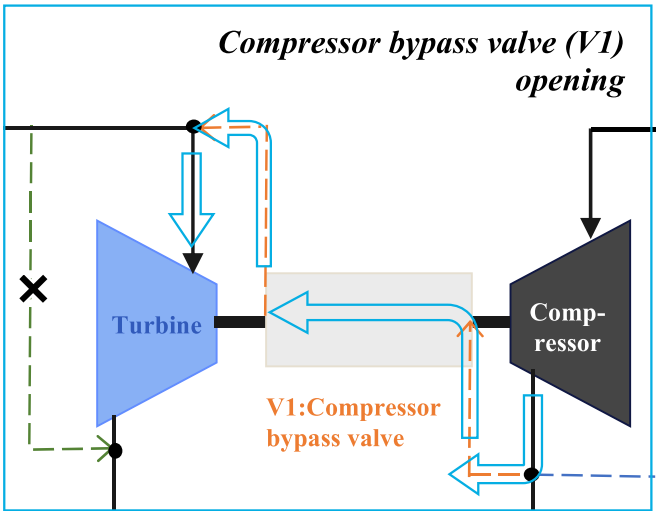


Fig. 5. Schematic diagram of the compressor bypass valve opening process.

Fig. 1 illustrates the structure and main components. The system uses a helium-xenon gas mixture as the working fluid, which is first heated in the reactor. Then, HeXe is expanded in the turbine, generating power, with both temperature and pressure decreasing. Afterward, it flows into the hot side of the recuperator and the precooler, where excess heat is removed. The cooled fluid enters the compressor, pressurized to the low-temperature and high-pressure fluid. It then passes through the recuperator to be heated and finally returns to the reactor inlet. The turbine, compressor, and generator are coaxially arranged. The design steady-state working condition of the system is shown in Table 3.

The system incorporates three types of naalves. The compressor bypass valve (V1) installed at the compressor outlet, allows low-temperature fluid to bypass the recuperator and reactor core, flowing directly to the turbine inlet. The turbine bypass valve (V2) located at the turbine inlet, enables high-temperature fluid to bypass the turbine and flow directly to the hot-side inlet of the recuperator. The compressor return valve (V3) installed at the compressor outlet, returns part of the fluid to the compressor inlet, increasing the mass flow rate through the compressor. Additionally, a storage tank is included, with its outlet valve connected to the compressor inlet, where the cycle pressure is the lowest, allowing adding fluid to the cycle using the pressure difference. The inlet valve is connected to the compressor outlet to remove the fluid using the lower pressure inside the tank. These valves, together with the tank, allow the system to effectively manage and adjust mass flow rate and pressure, ensuring safe operation and optimal performance.

2.2. Reactor core

The reactor core model consists of a neutron physics model and a thermo-hydraulic model. The neutron dynamics are described using the point kinetics equations (PKE), which are widely used and have been verified [44]. The PKEs are expressed as follows:

$$\frac{dn(t)}{dt} = \frac{\rho(t)-\beta}{\Lambda} n(t) + \sum_{i=1}^{6} \lambda_i C_i(t) \tag{1}$$

$$\frac{dC_i(t)}{dt} = \frac{\beta_i}{\Lambda} n(t) - \lambda_i C_i(t), i = 1, 2, \bullet\bullet\bullet, 6 \tag{2}$$

in the neutron physics model, the total reactivity is influenced by control mechanisms and the Doppler effect triggered by fuel temperature changes, as shown in equation (3). Other feedback effects, such as structural material temperature variations, are ignored.

$$\rho_{d,1} = 1 - (1-\rho_{d,ref}) e^{-\alpha_T \ln\left(\frac{T_1}{T_{ref}}\right)} \tag{3}$$

Where $\rho_{d,1}$ represents the reactivity caused by the current fuel temperature, ρ_{ref} is the referenced reactivity, T_1 represents the current fuel temperature, T_{ref} is the referenced fuel temperature, α_T represents the Doppler coefficient.

In the thermal-hydraulic model, the structure is simplified as shown in Fig. 2 and following assumptions are made [45]: (1) the heat source consists of fuel assemblies only; (2) there is only radial heat transfer within the reactor core, and axial heat transfer is ignored. (3) the power generated by fuel has a trigonometric distribution in the axial direction and this distribution remains constant during transient conditions.

2.3. Heat exchanger

The heat exchangers in the cycle include the recuperator and precooler, both of which are printed circuit heat exchangers (PCHE) arranged in countercurrent flow. A one-dimensional simulation method is used, dividing the PCHE into continuous control volumes along the flow direction. The calculation model and process are given in Fig. 3. Spatially, the outlet temperature of the previous node is used as the inlet temperature for the subsequent node. Temporally, a "quasi-steady state" approach is applied to improve speed. Although this calculation method will cause the update of the cold side one step later than that of the hot side, by selecting a reasonable time step, the error could be ignored [31].

Governing equations of heat exchangers are shown below with an example of the recuperator.

$$C_{p,c}\frac{dm_c T_{l,out}}{dt} = \dot{m} C_{p,c}\left(T_{c,in} - T_{c,out}\right) + h_{rcp-c} A_{rcp}\left(T_{rcp} - T_{c_{avr}}\right) \tag{4}$$

$$C_{p,h}\frac{dm_h T_{h,out}}{dt} = \dot{m} C_{p,h}\left(T_{h,in} - T_{h,out}\right) + h_{rcp-h} A_{rcp}\left(T_{rcp} - T_{h_{avr}}\right) \tag{5}$$

$$m_{rcp} C_{p,rcp}\frac{\partial T_{rcp}}{\partial t} = h_{rcp-h} A_{rcp}\left(T_{h_{avr}} - T_{rcp}\right) - h_{rcp-c} A_{rcp}\left(T_{rcp} - T_{c_{avr}}\right) \tag{6}$$

where index c stands for the cold-side flow, h stands for the hot-side flow. The physical parameters of HeXe are calculated using the average temperature within each node.

2.4. Turbomachinery

The turbomachinery in the cycle is the turbine, the compressor, and the alternator, rotating at the same speed. In this paper, the grid connection mode for TAC is synchronous mode, with the shaft speed fixed by the grid frequency. In this mode, Equation (7) is used for calculating the output power.

$$P_{alt} = P_{tur} - P_{com} \tag{7}$$

$$P_{tur} = \dot{m} C_{p,T_{t_{avr}}}\left(T_{t,in} - T_{t,out}\right) \tag{8}$$

$$P_{com} = \dot{m} C_{p,T_{c_{avr}}}\left(T_{c,out} - T_{c,in}\right) \tag{9}$$

Outlet parameters of the turbine and the compressor are derived from the temperature ratios (TR) and pressure ratios (PR), which can be calculated from the characteristic curves, given in the appendix. In this

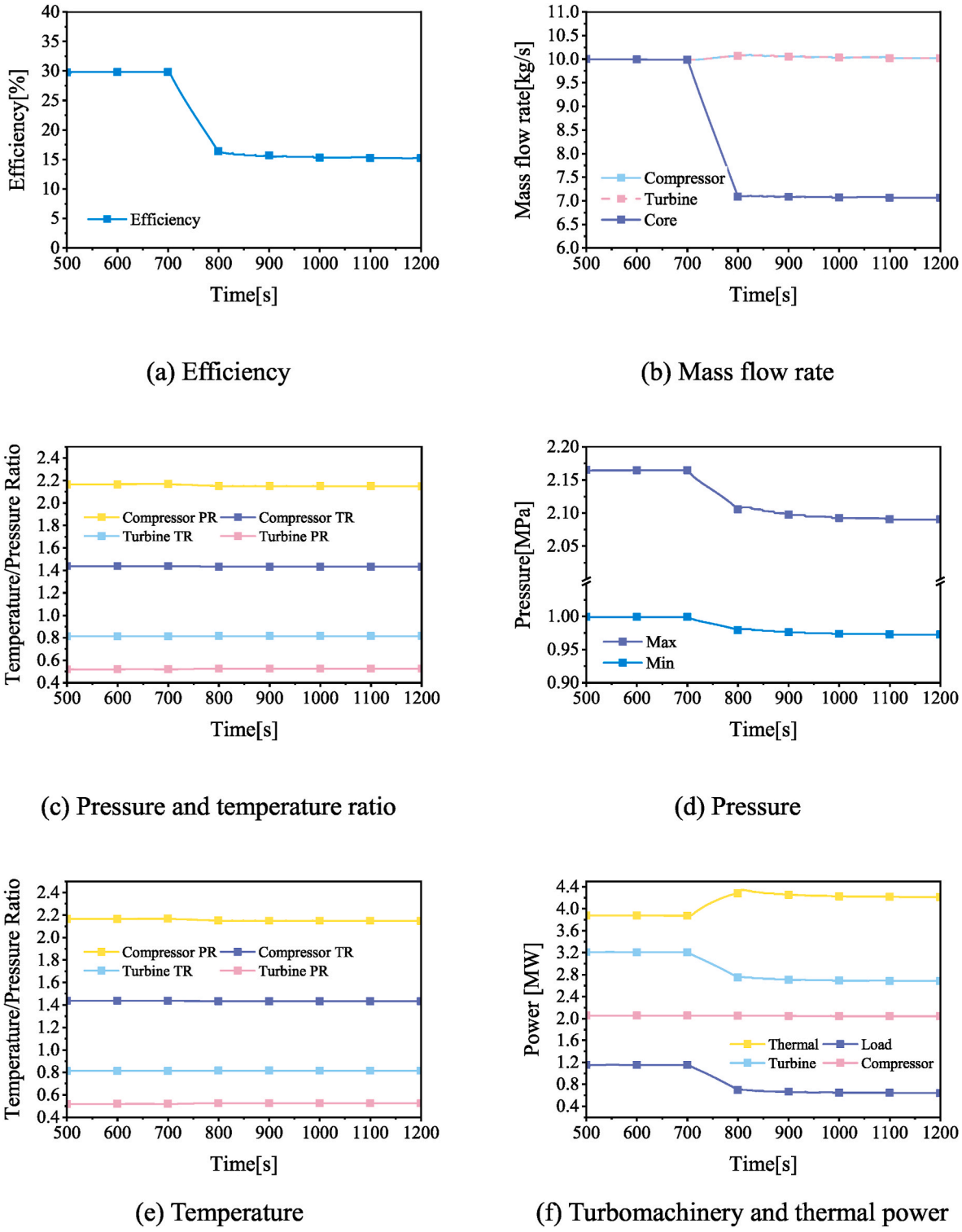


Fig. 6. Open loop simulation curve under the compressor bypass valve.

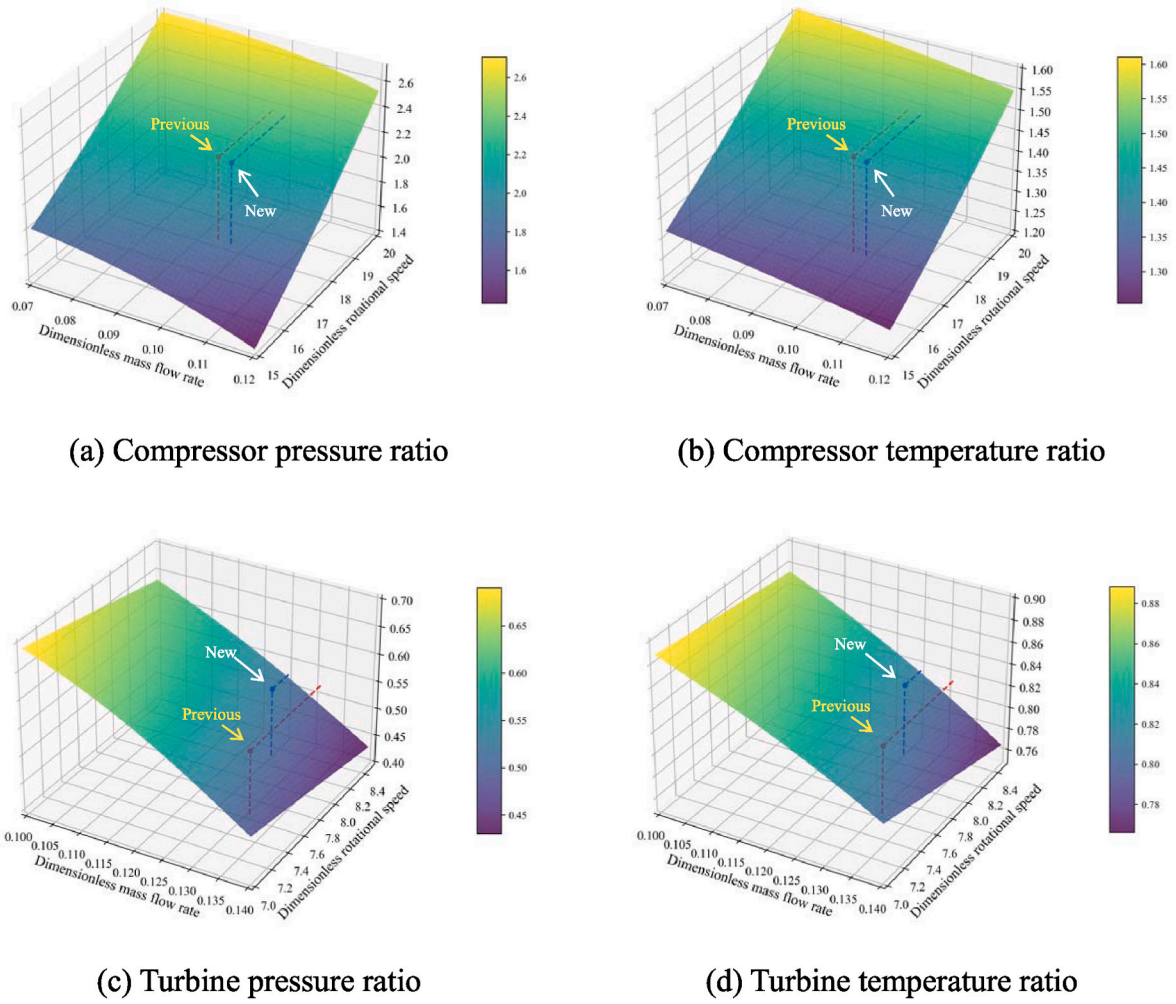


Fig. 7. Variation of state point in turbomachinery performance map under the compressor bypass valve.

study, the curves have been fitted into polynomials on the dimensionless mass flow rate and the dimensionless rotating speed [27]. These parameters can be calculated using Equation (10) and Equation (11) [46].

$$m' = \frac{\dot{m}\sqrt{\frac{T_{in}R_0}{\gamma}}}{(2r_{tip})^2 P_{in}} \tag{10}$$

$$N' = \frac{N \cdot 2r_{tip}}{\sqrt{\gamma T_{in} R_o}} \tag{11}$$

To ensure safe operation, the surge margin of the compressor is calculated using Equation (12), which is expected to be greater than 20 % throughout the process.

$$SM = \left(\frac{PR_s^*}{PR} \times \frac{\dot{m}}{\dot{m}_s} - 1\right) \times 100\% \tag{12}$$

2.5. Valves

In this study, valves are modeled as linear valves, with flow area linearly related to their opening. For the linear valve model, when the differential pressure between the inlet and outlet is large, the following equation can be used [47]:

$$\dot{m} = \begin{cases} C_d f_{open} A \sqrt{\frac{2\gamma}{\gamma-1} P_{in}\rho_{in}\left[\left(\frac{P_{out}}{P_{in}}\right)^{\frac{2}{\gamma}} - \left(\frac{P_{out}}{P_{in}}\right)^{\frac{\gamma+1}{\gamma}}\right]}, \\ \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}} \le \frac{P_{out}}{P_{in}} \le 1 \\ C_d f_{open} A \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2(\gamma-1)}} \sqrt{\gamma P_{in}\rho_{in}}, \\ 0 \le \frac{P_{out}}{P_{in}} \le \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}} \end{cases} \tag{13}$$

where, f_{open} represents the ratio of valve opening, C_d is the valve discharge coefficient, defined as:

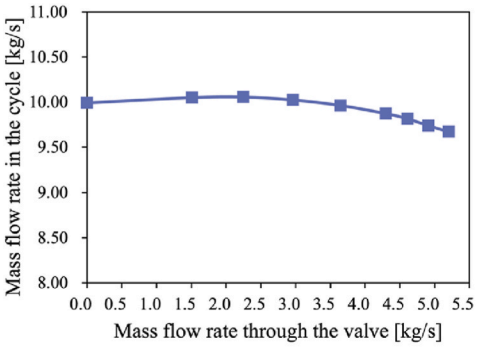
$$C_d = 0.0112e^{0.196\frac{L}{D}} \tag{14}$$

where, L means valve Lift, D is its diameter.

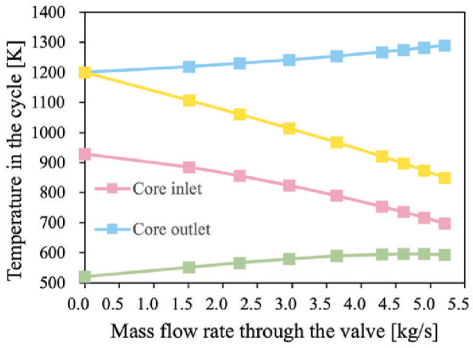
2.6. The storage tank

In this study, assuming that the storage tank is not connected to any external heater, the conservation of mass and energy is described by Equations (15) and (16) [29]:

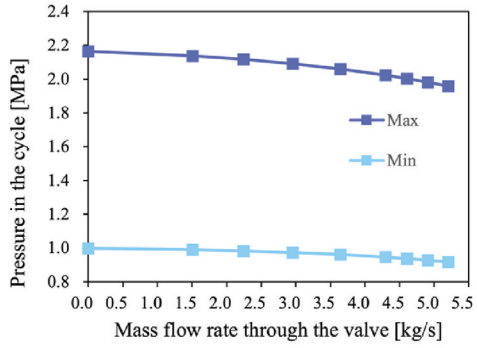
$$\frac{dM_{tank}}{dt} = \dot{m}_{in} - \dot{m}_{out} \tag{15}$$



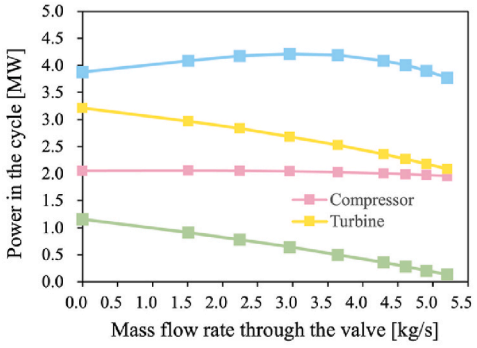
(a) Mass flow rate



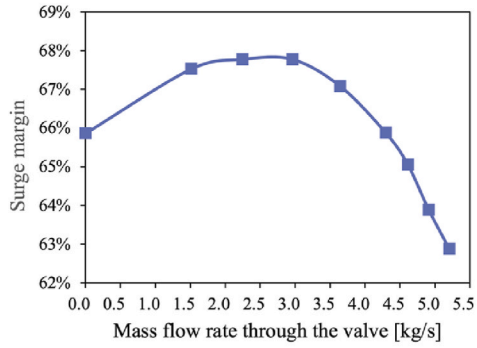
(b) Temperature



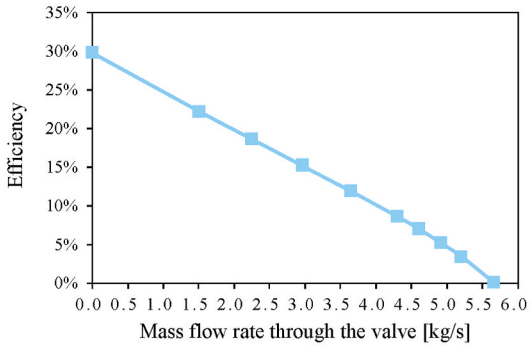
(c) Pressure



(d) Turbomachinery and thermal power



(e) Compressor surge margin



(f) Efficiency

Fig. 8. Changes in parameters under the compressor bypass control.

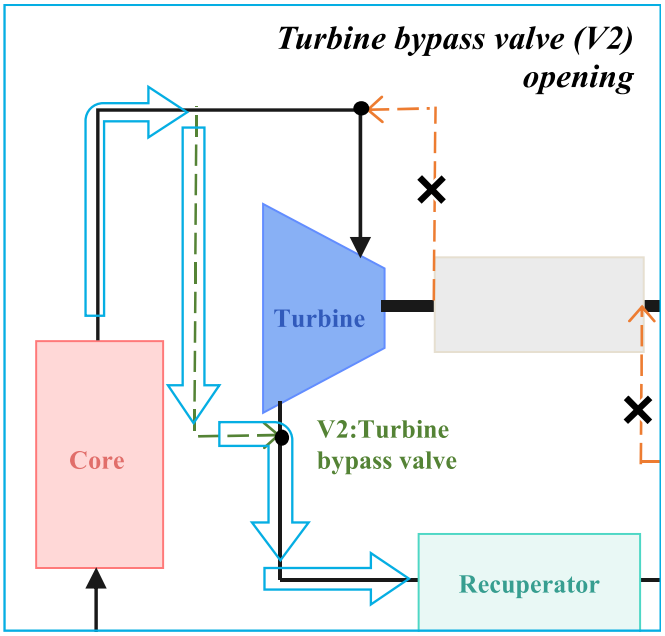


Fig. 9. Schematic diagram of the turbine bypass valve opening process.

$$\frac{d(M_{tank}h_{tank})}{dt} = \dot{m}_{in}h_{in} - \dot{m}_{out}h_{tank} + V_{tank}\frac{dP_{tank}}{dt} \tag{16}$$

where, h_{tank} represents the enthalpy of the working fluid in the tank. Considering the relationship between physical properties, following equations could be get:

$$\frac{dh_{tank}}{dt} = \frac{1}{M_{tank} - V_{tank}\frac{dP_{tank}}{dh_{tank}}\Big|_{\rho_{tank}}}\left[\dot{m}_{in}(h_{in} - h_{tank}) + \frac{dP_{tank}}{d\rho_{tank}}\Big|_{h_{tank}}(\dot{m}_{in} - \dot{m}_{out})\right] \tag{17}$$

2.7. Conservation of mass

For a closed Brayton cycle, the total mass of the fluid in the system remains constant. Specifically, the sum of the mass in the storage tank and the cycle remains constant, as expressed in the following equation:

$$M_{sys} = M_{cycle} + M_{tank} = constant \tag{18}$$

where the mass of the fluid in the cycle equals to the sum of the inventory in every individual component. At every time step, the following Equation serves as a restriction, determining the mass flow rate in the cycle, and is solved together with the pressure.

$$M_{fill} = \sum_{n}\frac{p_i V_i}{R_0 T_i} \tag{19}$$

3. Code and verification

In this study, the dynamic model is established using Python, with both steady-state and transient process calculations performed. The system's governing ordinary differential equations (ODEs), as described in Section 2, are solved using "scipy.integrate" library [48], which is widely recognized for its reliability and effectiveness. To evaluate the code's accuracy, the key parameters under full power conditions are compared with the design values, showing pretty consistency, as shown in Table 4.

For further verification, the code in this paper is used to simulate the SNRP system from the article [27], which also employs HeXe as the working fluid and integrates the reactor with CBC for energy conversion. The boundary conditions, initial conditions, and design parameters in this verification are closely matched to those in Ref. [27] to ensure reliability and accuracy. The verification results in Fig. 4, display the curves of key parameters in the start-up transient process, showing good agreement.

4. Results and discuss

To enable variable power output, the study adjusts the compressor bypass valve, turbine bypass valve, and the amount of working fluid in the cycle. Additionally, compressor anti-surge control is employed to maintain a surge margin greater than 20 %, and compressor inlet temperature (CIT) control is used.

Dynamic simulation for control strategies is categorized into open-loop and closed-loop simulations [47]. In open-loop simulations, the input signal remains unaffected, with no feedback from the simulation results. This type of simulation for three methods is first studied in Section 4.1. Subsequently, closed-loop simulations, where the input signal is influenced by the output signal, like most control systems operate [47], are conducted. In Section 4.2, three different load-following methods are compared based on their characteristics, such as their ability to achieve fast, fluctuation-free load-following while maintaining system efficiency, and their effects on the system safety. Finally, in Section 4.3, step and ramp changes are used to demonstrate the effectiveness of these control strategies.

4.1. Open-loop simulation

In this section, open-loop simulations for compressor bypass valve, turbine bypass valve and inventory control are described in turn, which is fundamental for studying the dynamic behavior of the system.

4.1.1. Compressor bypass valve

Fig. 5 shows the process of the compressor bypass valve (V1) opening. The transient process, in which the opening of V1 changes from 0 % to 40 %, is used to analyze the mechanism and characteristics of this method.

The simulation results are shown in Fig. 6. The system starts in full power condition. Then, from 700 to 800 s, V1 is gradually opened as shown in Fig. 6a. As the valve opens, the mass flow rate through the core decreases significantly, while a slight increase occurs at the turbine and compressor inlets, as seen in Fig. 6b. The performance of the turbomachinery, which is critical to the cycle's characteristics, is influenced by dimensionless parameters related to mass flow rate, rotational speed, inlet temperature, and inlet pressure, as defined in Equations (10) and (11). As shown in Fig. 7, the TR and PR remain nearly unchanged despite variations in these parameters, consistent with Fig. 6c. The expansion of the flow path decreases the pressure in the cycle. The pressure on both the high- and low-pressure sides decreases in approximately equal proportion, from 2.17 MPa to 2.09 MPa and 1.00 MPa–0.97 MPa, respectively, as shown in Fig. 6d.

Due to the existence of CIT control, the compressor inlet temperature does not change, nor does the temperature at the recuperator cold-side inlet, with the essentially unchanged TR of the compressor. However, the reduced mass flow rate at the recuperator's cold side causes a temperature drop at the cold-side outlet. Meanwhile, the hot side fluid is

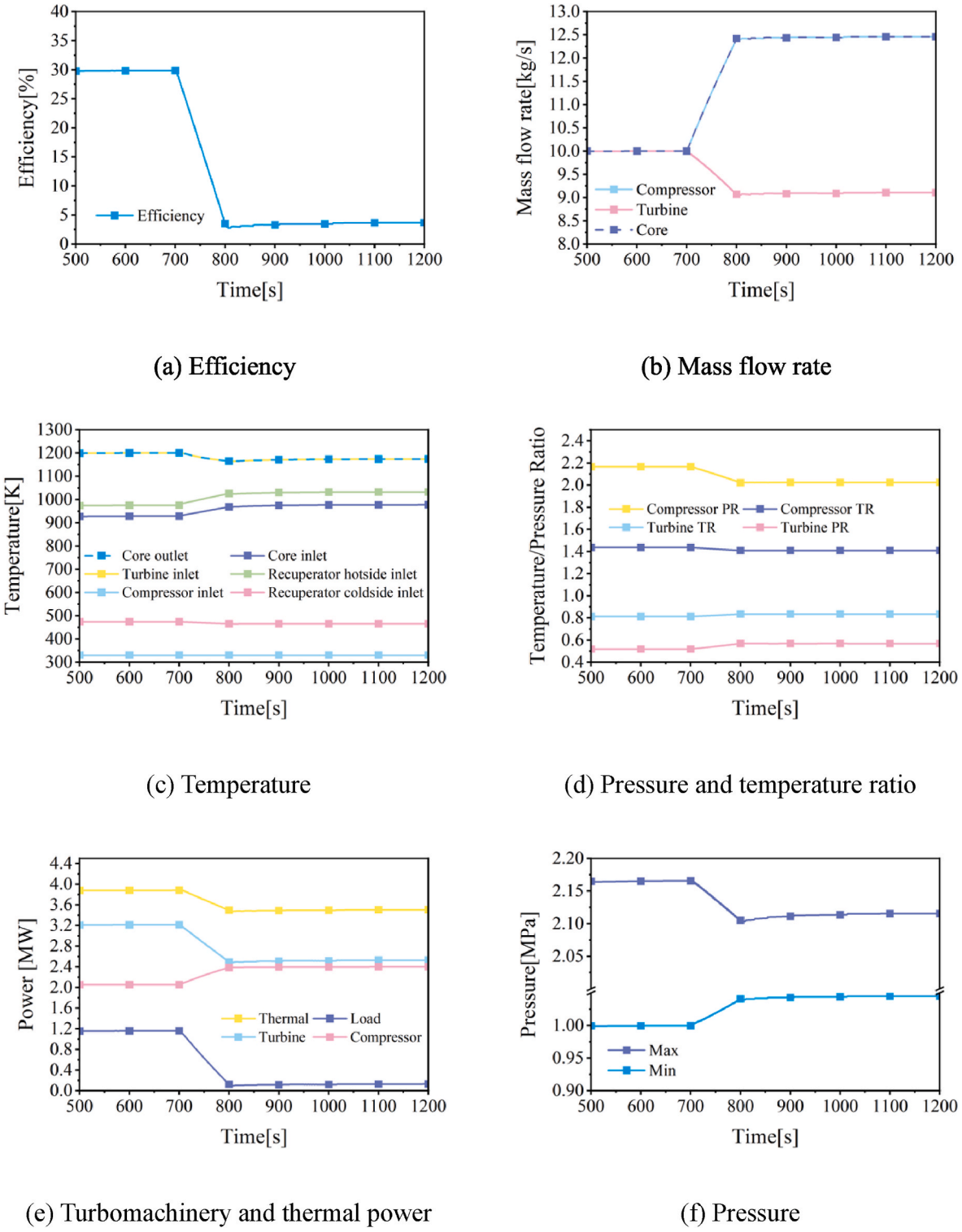


Fig. 10. Open loop simulation curve under the turbine bypass control.

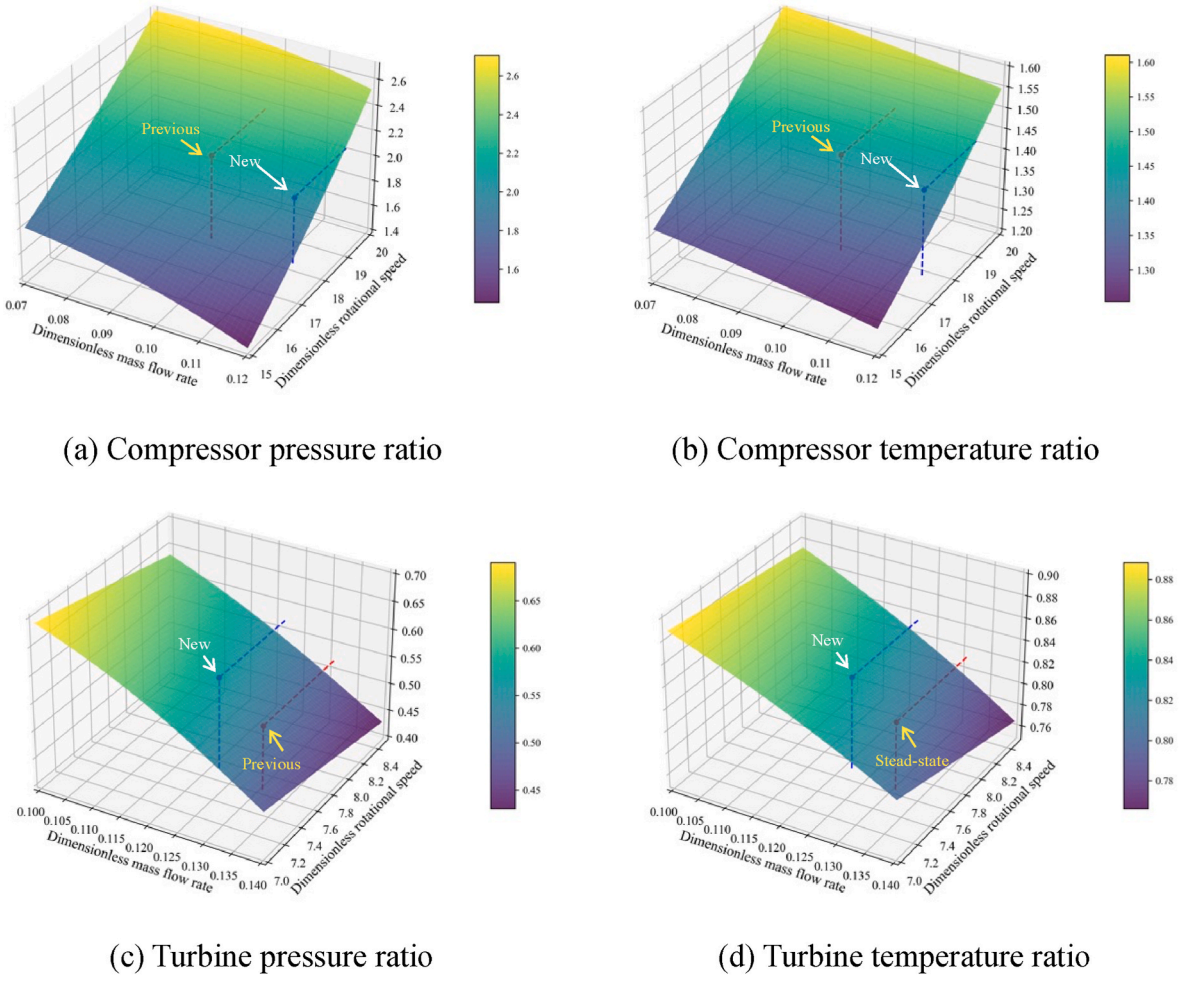


Fig. 11. Variation of state point in turbomachinery performance map under the turbine bypass valve.

insufficiently cooled, increasing the precooler inlet temperature. Therefore, to maintain a constant precooler outlet temperature, its cooling ability needs to meet a higher demand.

Considering the energy exchange in the reactor, when the balanced situation is reached, all the heat generated will be removed by the coolant, as described by Equation (20).

$$P_{core} = \dot{m} C_{p,T_{avr}} (T_{in} - T_{out}) \tag{20}$$

Therefore, with a decrease in mass flow rate and an increase in reactor power, the core outlet temperature rises from 1200 K to 1242 K. Besides, the turbine inlet temperature is also directly affected. As cold fluid from the compressor outlet flows directly to the turbine inlet, the temperature decreases. Rewriting Equation (8), the work done by the turbine is given.

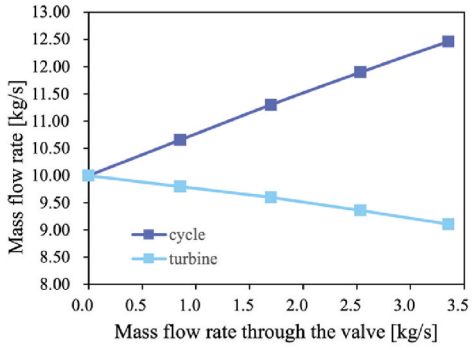
$$P_{tur} = \dot{m} C_{p,T_{avr}} T_{t,in} (1 - TR_{tur}) \tag{21}$$

So, its inlet temperature decrease leads to a decrease in its produced power, as shown in Fig. 6f. Meanwhile, the work consumed by the compressor remains basically the same.

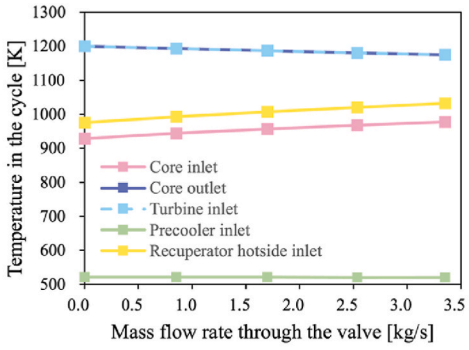
Since the change in mass flow rate through V1 affects system performance, the curves of key parameters varying with it are shown in Fig. 8. With the increase in mass flow rate through the valve, the mass flow rate in the cycle initially increases slightly, then decreases due to the combined effects of flow path expansion, pressure drop, and temperature-induced density changes. The compressor surge margin follows a similar trend, showing no risk. Regarding temperature, the highest value is observed at the core outlet. When the system output is adjusted to 0 MWe, the temperature increases to 1302.62K, within the limit. Simultaneously, the requirement for the precooler cooling capacity rises by 37.46 % compared to the full-power condition. And there is no overpressure risk. From the previous analysis, this method mainly alters the system output power by reducing turbine work, while the compressor's consumed work remains largely unchanged, consistent with Fig. 8d. Also, as the core's thermal power increases slightly, system efficiency decreases, as shown in Fig. 8f.

4.1.2. *Turbine bypass valve*

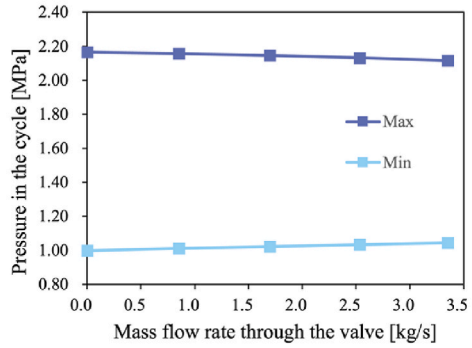
Fig. 9 depicts the process of the turbine bypass valve (V2) opening. The simulation results as the opening of V2 changes from 0 % to 40 % are shown in Fig. 10. With V2 turned on, the mass flow rate through the reactor and compressor increases significantly, while that through the turbine decreases, as shown in Fig. 10b. The performance map's state points also shift, as shown in Fig. 11, contributing to changes in PRs. It results in a reduced pressure difference between the high-pressure and low-pressure sides, with the minimum pressure increasing from 1 MPa to 1.05 MPa, and the maximum pressure decreasing from 2.16 MPa to 2.11



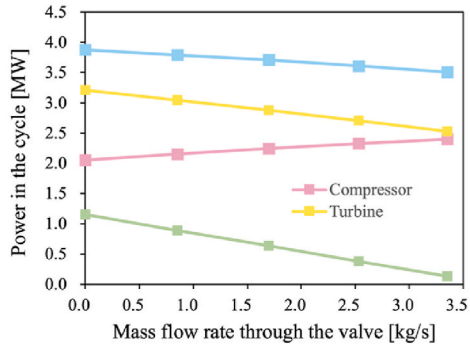
(a) Mass flow rate



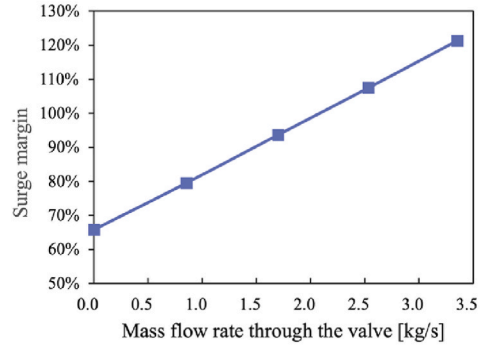
(b) Temperature



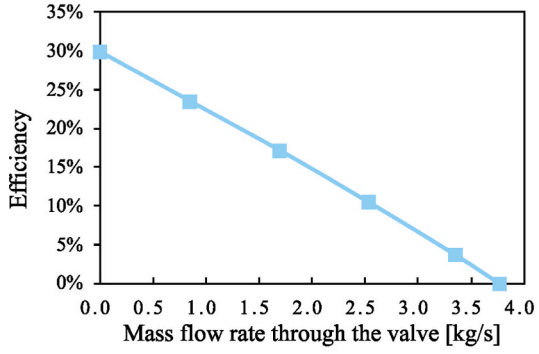
(c) Pressure



(d) Turbomachinery and thermal power



(e) Compressor surge margin



(f) Efficiency

Fig. 12. Changes in parameters under the turbine bypass control.

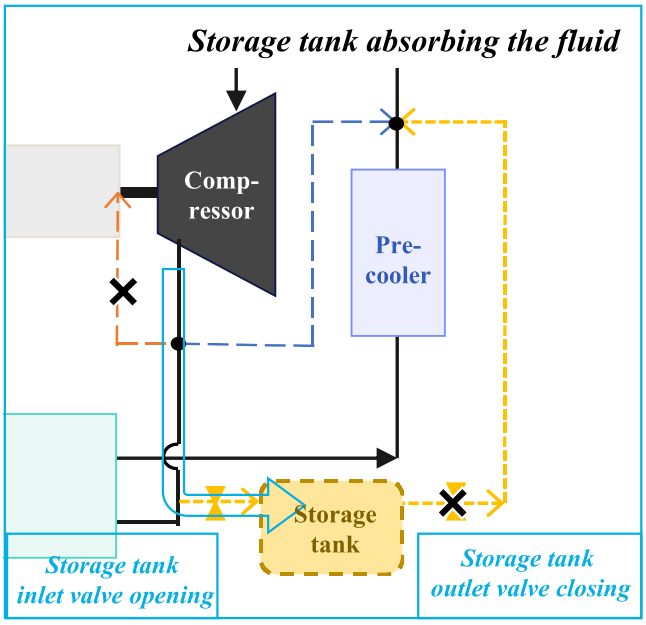


Fig. 13. Schematic diagram of the storage tank absorbing the working fluid from the cycle.

MPa, as shown in Fig. 10f.

Due to the CIT control and unchanged compressor TR, the temperatures at the compressor inlet and recuperator cold-side inlet remain nearly constant. Meanwhile, the mass flow rate on both sides of the recuperator stays the same, causing the cold-side outlet temperature to rise as the hot-side inlet temperature increases. As the reactor inlet temperature continues to rise, the fluid's cooling ability weakens, leading to reduced heat exchange within the reactor. This causes a decrease in reactor fission power due to the Doppler effect, as shown in Fig. 10e. According to Equation (21), the turbine mass flow rate and the inlet temperature decrease, resulting in a slight increase in TR and a reduction in turbine work. For the compressor, as its mass flow rate increases, so does its consumed work.

To conclude, curves of key parameters varying with the mass flow rate through V2 are obtained in Fig. 12. As the mass flow rate through the valve increases, the mass flow rate through the turbine decreases, while others in the cycle increases. Regarding temperature, there is an increase in the recuperator, requiring it to withstand higher-temperature fluid, about 106 % of the design value. Moreover, this method reduces output power by decreasing turbine work and increasing compressor power consumption simultaneously. In this process, the thermal power also decreases, but less than the output power, leading to a decrease in system efficiency. Additionally, the opening of V2 does not pose safety issues on overpressure or compressor surge.

4.1.3. Inventory

Another method is inventory control. While two types of valve control described above offer solutions for meeting rapidly changing load demands, they result in lower system efficiency. Inventory control, on the contrary, has a smaller impact on system efficiency. It adjusts the output power by regulating the mass of the fluid in the cycle, which in turn affects the mass flow rate and pressure [34], with its process shown in Fig. 13.

In this simulation, the system operates at full power until 400 s. At 400 s, the inlet valve of the storage tank opens, beginning to absorb the fluid from the cycle. As a result, the mass in the cycle decreases, leading to a corresponding decrease in the mass flow rate, as shown in Fig. 14a and b. This also causes a decrease in the surge margin, as depicted in Fig. 14h. Simultaneously, the pressure in all positions of the cycle drops rapidly, while the one in the tank rises, as shown in Fig. 14f.

Around 1050 s, when output power drops to 0.93MWe, about 80 % of full power, the compressor's surge margin reaches the minimum allowable value of 20 %, triggering anti-surge protection. Consequently, the compressor return valve opens, preventing a further decrease in the mass flow rate through the compressor, as shown in Fig. 14b. System efficiency remains unchanged until the return valve opens but declines significantly afterward, as indicated in Fig. 14g. This is because the opening of the return valve does not affect the rate at which the power produced by the turbine declines but significantly impacts the power consumed by the compressor, maintaining it at a steady level. This leads to an increase in the decreasing speed of the output power, while the decreased speed of thermal power does not change, as illustrated in Fig. 14e.

At 1400 s, the highest pressure in the cycle equalizes with the pressure in the tank, as shown in Fig. 14f. With no differential pressure to drive the flow, the tank stops absorbing the working fluid, and the system gradually stabilizes into a new steady state. At that time, the output power is about 30 % of full power.

In terms of temperature, the compressor inlet temperature does not remain at 330 K throughout the process, unlike the valve control methods. This is because after the return valve opens, it brings in higher-temperature fluid without cooling.

Key parameters varying with the mass in the cycle are shown in Fig. 15. Under inventory control, the temperature in the cycle increases, but the risk of over-temperature is low, with a maximum temperature of 1270 K. There is no overpressure risk, but the decrease in mass flow rate may trigger compressor surge. Therefore, inventory control should be used in conjunction with anti-surge control. However, the anti-surge control greatly influences the characteristics of the system. According to this, the process can be divided into three stages.

1) First, the reduction of the mass flow rate in the cycle has not triggered the anti-surge control and system efficiency does not show a decrease.
2) Second, when the output power reaches 80 %, the return valve opens, reducing cycle efficiency by making the compressor consumed work no longer reduced.
3) Third, when the highest pressure in the system drops to the same level as the internal pressure of the tank, the mass in the cycle is no longer reduced, thus this method no longer works. At that time, the output power is about 30 % of full power.

4.2. Closed-loop simulation

In this section, closed-loop simulations are performed to achieve load following using a PID controller. The PID parameters of control methods used in this study are listed in the appendix. The valve control diagram, shown in Fig. 16, covers both the compressor bypass valve and the turbine bypass valve.

Previous studies have indicated that inventory control suffers from slower response times and higher fluctuations in closed-loop simulations [49]. This is due to the inherent characteristics of the tank, which operates as a pressure-driven process. Thus, it is difficult to achieve optimal results by just adjusting the PID parameters. Therefore, an optimized inventory control method, combined with the bypass valve, is necessary. All inventory controls mentioned below refer to this

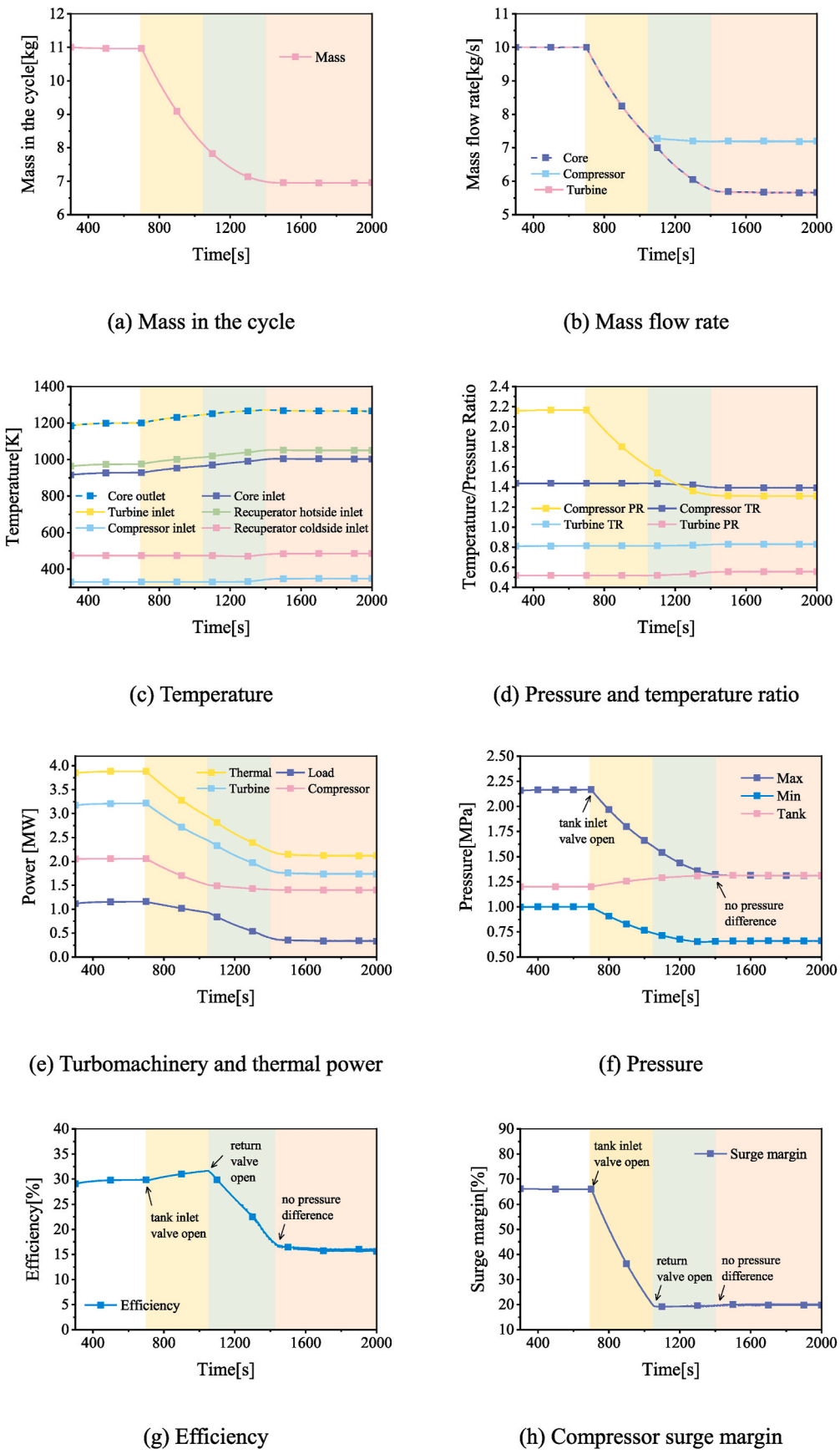


Fig. 14. Open loop simulation curve under inventory control.

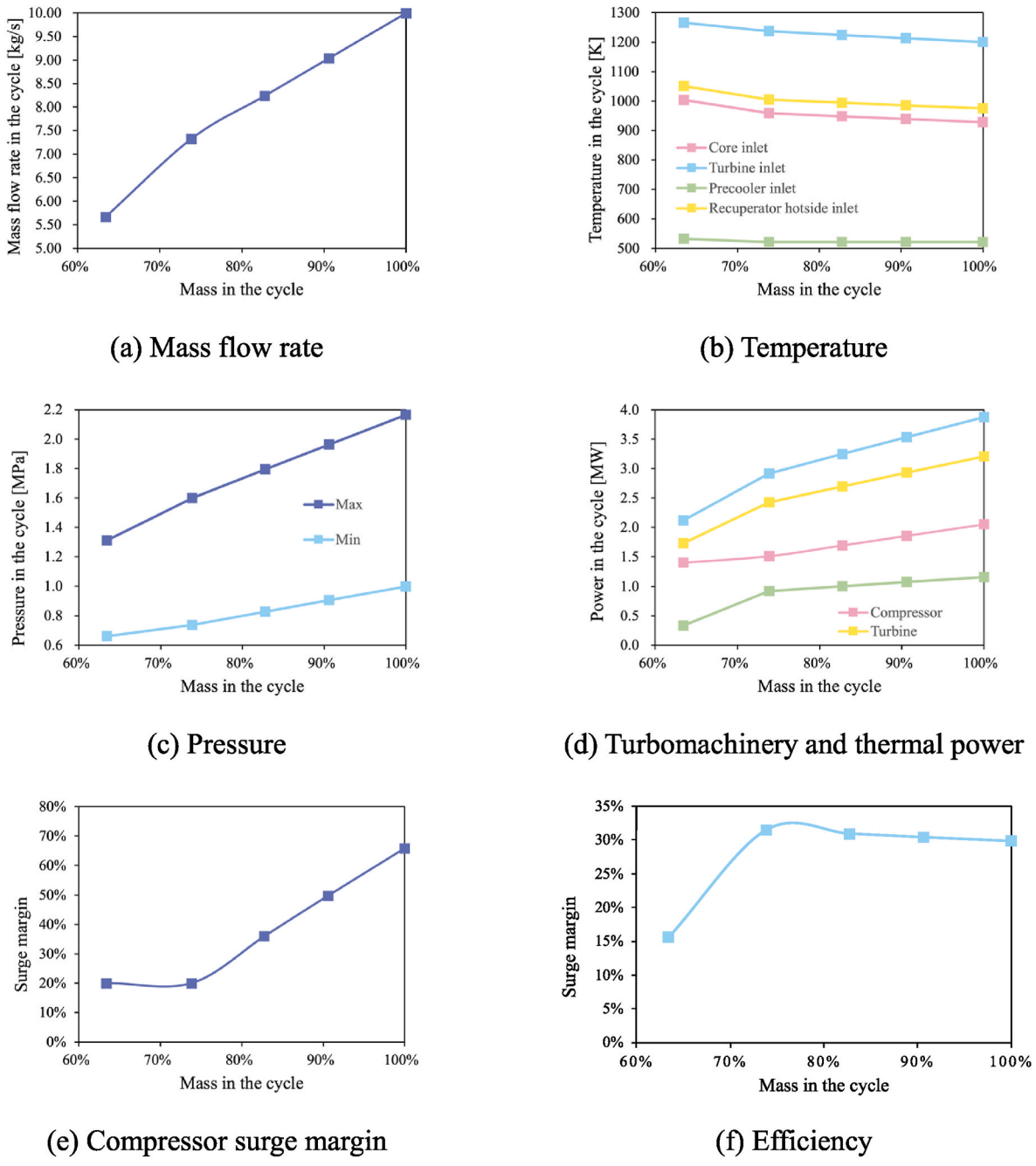


Fig. 15. Changes in parameters under inventory control.

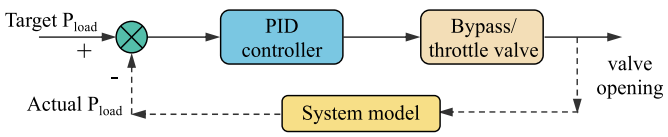


Fig. 16. Valve load-following control PID diagram.

optimized method. This method is described as follows.

1) First, obtain the "corresponding value" of mass in the cycle for a given output power.
2) In practice, according to the target output power, use the correspondence to operate the tank valves. Simultaneously, the bypass valve is utilized for fast adjustments to compensate for the initial slow response. Once the tank completes its regulation of mass in the cycle, the bypass valve will automatically close.

Furthermore, as seen in Section 4.1, inventory control exhibits three distinct characteristics during the entire process. Therefore, in this section, the transient process, in which the output power decreases from 100 % to 0 %, is divided into three stages. Closed-loop simulations are carried out for each of the three control methods to compare their characteristics and impacts on the key parameters of the system.

4.2.1. *First stage:100%–80 %*

In this section, three control methods are utilized to achieve the transient process where the output power changes from 100 % to 80 % in 100 s. The variation curves of key parameters are shown in Fig. 17 and the steady-state values of them at the 80 % output power under each

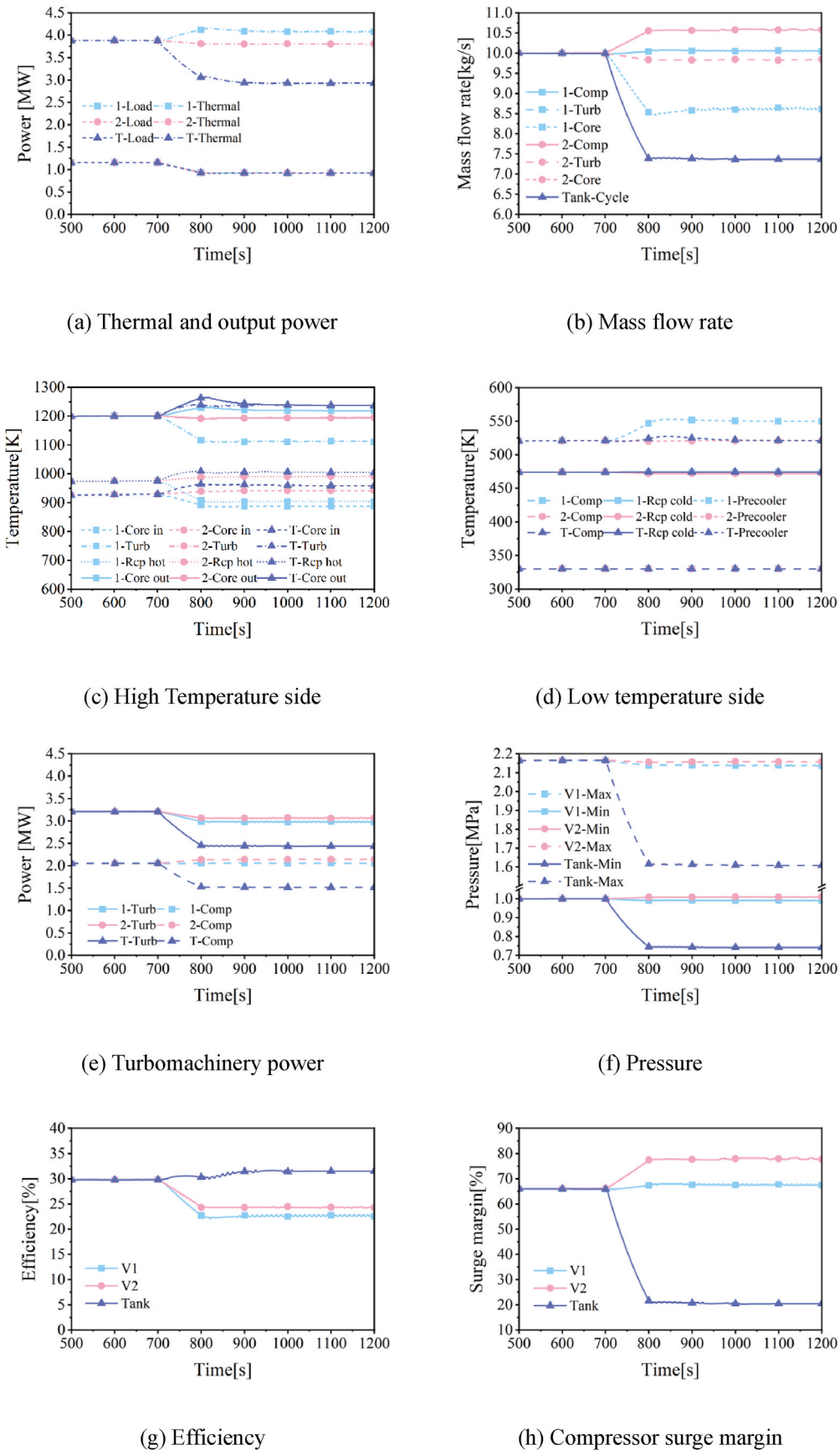


Fig. 17. Close-loop simulation for three control methods with output power from 100 % to 80 %.

Table 5
Parameters at 80 % output power of three control methods.

	design working condition	V1(compressor bypass valve)	V2(turbine bypass valve)	Tank(Inventory control)
Max Temperature[K]	1200	1218.42	1194.35	1236.70
Min Temperature[K]	330.00	330.00	330.00	330.00
Max Pressure[MPa]	2.17	2.14	2.16	1.61
Min Pressure[MPa]	1	0.99	1	0.74
Mass flow rate [kg/s]	10	10.05	10.58	7.37
Surge margin[%]	65.89 %	67.44 %	77.65 %	20.48 %
Load power[MW]	100 %	80 %	80 %	80 %
System efficiency[%]	29.77 %	22.54 %	24.30 %	31.08 %

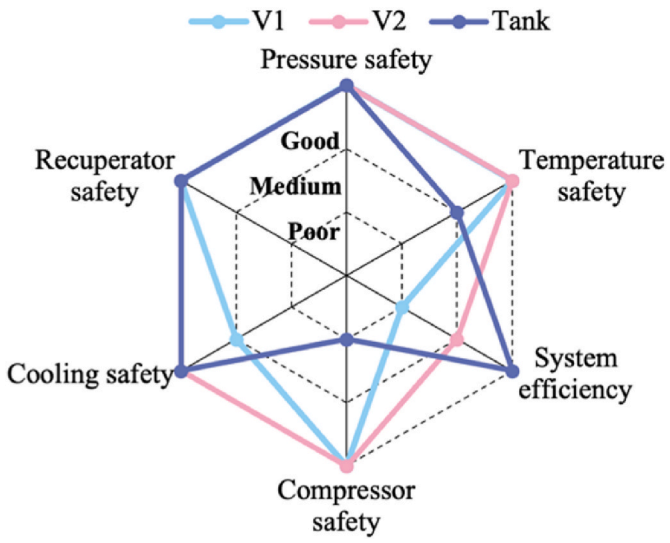


Fig. 18. Performance comparison of three control methods in the process with output power from 100 % to 80 %.

method are presented in Table 5. As shown in Fig. 17a, the output power under all three control strategies ramps down from 600 s, closely matching the target. All three methods achieve load following with almost no offset or fluctuation.

As shown in Fig. 17b, V2 control has the least impact on the mass flow rate, with the mass flow rate through the core and compressor rising to 10.58 kg/s, while the one through the turbine shows a slight decrease. V1 control causes the mass flow rate through the core to drop to 8.60 kg/s, while the one through the turbine and compressor rises slightly. Under inventory control, the mass flow rate of the entire cycle drops from 10 kg/s to 7.37 kg/s, a much greater change than under valve controls.

Regarding temperature, V2 control has the smallest overall impact, while inventory control causes an increase in temperature on the high-temperature side. The temperature change under V1 control is more complicated. On the high-temperature side, both V1 control and inventory control cause an increase in the maximum temperature of the cycle. Under inventory control, the core outlet temperature reaches 1264.74 K during the transient process, showing a more significant increase. Additionally, as the optimized inventory control initially involves V1, the temperature at the turbine inlet is lower than that at the core outlet early in the process, presenting similar characteristics to V1 control early in the process, but later aligning with inventory control characteristics. On the low-temperature side, the minimum temperature under all three methods is maintained at 330 K. However, for the precooler inlet temperature under V1 control, there is a significant increase from 520 K to 550 K, consistent with previous analyses, indicating that this method requires a better capability of removing waste heat. Under inventory control, temperature at the precooler inlet is first increased and followed by a decline, also due to the involvement of V1.

Regarding other parameters, inventory control exhibits distinct characteristics compared to valve controls. In terms of pressure, inventory control reduces the output power by decreasing the mass in the cycle, reducing the maximum and minimum pressures to 1.61 MPa and 0.74 MPa, respectively. In contrast, pressure changes under valve controls are more moderate. Additionally, only under the inventory control does the surge margin decrease from 66 % to 20 %, while for the others, it increases, consistent with the changes in the mass flow rate at the compressor inlet. Since anti-surge control was not implemented, the system efficiency under inventory control did not degrade but rather increased slightly from 30 % to 31.5 %, as shown in Fig. 17g. From Fig. 17a, the thermal power decreases the most under inventory control, contributing to its feature of the highest system efficiency.

At this stage, the characteristics of the three control methods are summarized in Fig. 18. None of them present a risk of overpressure or damage to the recuperator. Regarding waste heat discharge at the precooler and temperatures in the cycle, there is a 15.8 % increase in cooling capacity demand under V1 method, and a 5.4 % increase in the maximum temperature under inventory control. However, both risks remain marginal. The highest system efficiency of 31.08 % is achieved under inventory control, while the system efficiency drops to 22.54 % and 24.30 % for the V1 and V2 control methods, respectively. Although the surge margin drops to its lowest value under inventory control, it can still be considered the preferred regulation method for load following at this stage, provided there is reliable anti-surge control.

4.2.2. *Second stage:80%–30 %*

The variation curves of the key parameters during closed-loop simulations of load following from 80 % to 30 % are shown in Fig. 19. The steady-state values of key parameters at 30 % output power under each method are shown in Table 6. Before 1000 s, the system starts in a steady-state condition, regulated to 80 % output power using inventory control. As shown in Fig. 19a, all control methods achieve fast and fluctuation-free load following. Regarding mass flow rate, the methods exhibit characteristics like those described in Section 4.2.1. The highest temperature occurs during the regulation process under V1 method, with the core outlet temperature rising to 1330.18K and then gradually decreasing, stabilizing at 1291.68K. Concurrently, the temperature at the turbine and core inlet under the V1 method decreases more drastically, falling to 956.98K and 781.01K. On the low-temperature side, the precooler inlet temperature under the V1 method increases from 521.04K to 598.82K. Meanwhile, the compressor inlet temperature under inventory control rises due to the opening of the return valve. Due to the initially low mass flow rate through the return valve, the inlet temperature does not show a significant increase until around 1160 s.

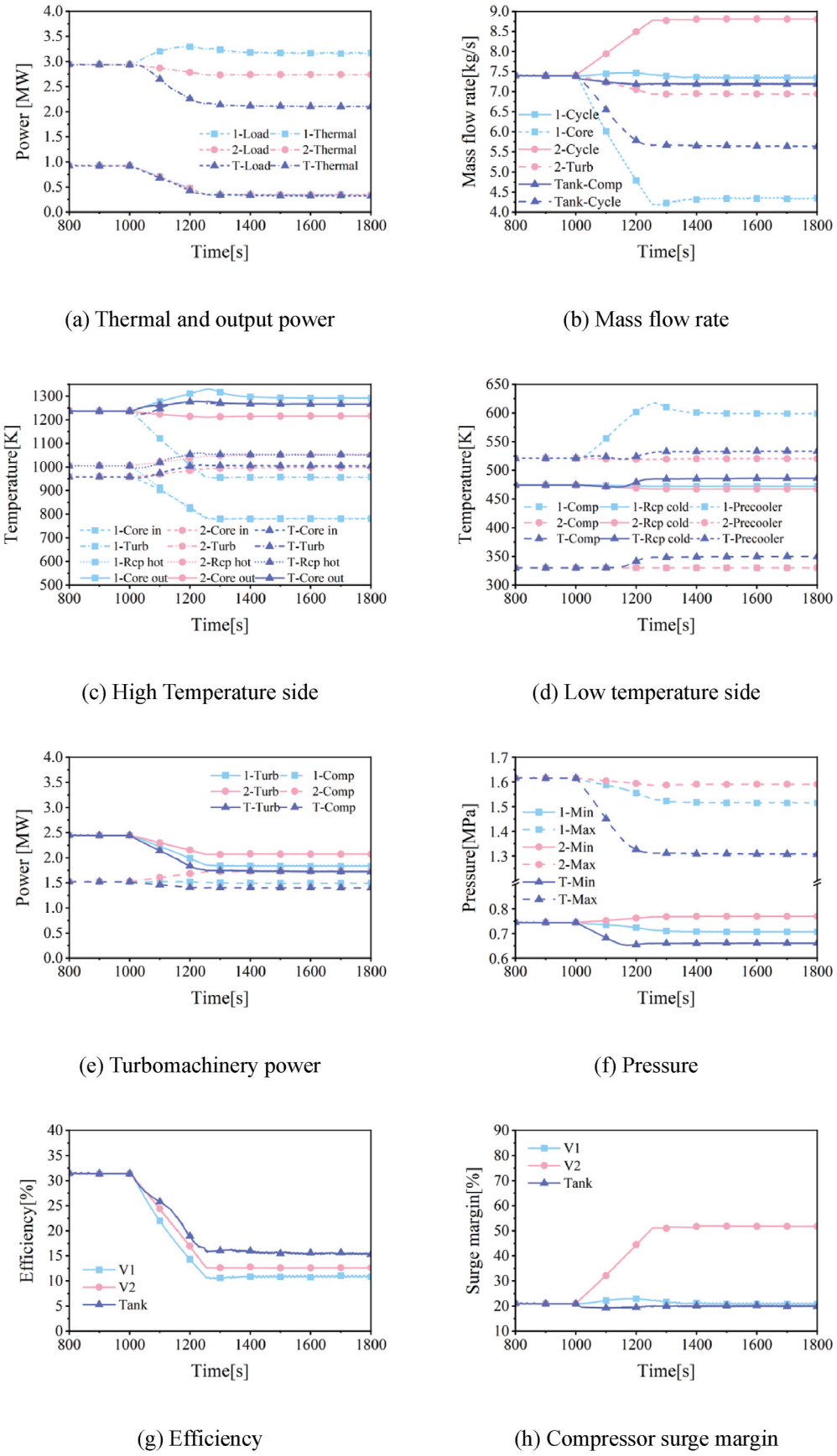


Fig. 19. Close-loop simulation for three control methods with output power from 80 % to 30 %.

Table 6
Parameters at 30 % output power of three control methods.

	80 % output power condition	V1(compressor bypass valve)	V2(turbine bypass valve)	Tank(Inventory control)
Max Temperature[K]	1236.7	1291.68	1215.91	1266.01
Min Temperature[K]	330	329.99	330.00	349.92
Max Pressure[MPa]	1.61	1.52	1.59	1.31
Min Pressure[MPa]	0.74	0.71	0.77	0.66
Mass flow rate[kg/s]	7.37	7.34	8.81	7.18
Surge margin[%]	20.48 %	20.82 %	51.78 %	19.84 %
Load power[MW]	80 %	30 %	30 %	30 %
System efficiency[%]	31.08 %	10.84 %	12.62 %	15.33 %

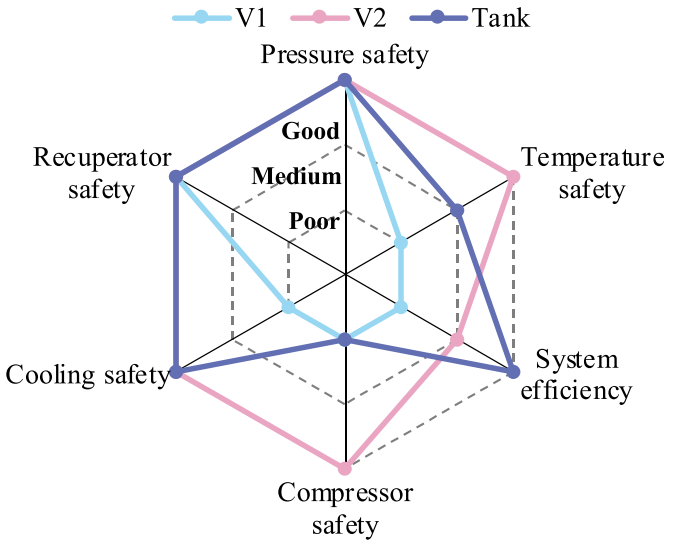


Fig. 20. Performance comparison of three control methods in the process with output power from 80 % to 30 %.

Overall, the temperature variation under V2 method is minimal.

Regarding pressure, the change is most drastic under inventory control. In the second stage, the use of the return valve under inventory control leads to a decrease in system efficiency. However, the drop in thermal power is the most significant, resulting in a relatively higher system efficiency compared to the two valve control methods, as shown in Fig. 19g. Additionally, during this stage, the compressor surge margin under the V1 method decreases to the minimum allowable value. In contrast, it increases from 20 % to 51.78 % under the V2 method.

Fig. 20 summarizes the safety characteristics and efficiency of the system during the load-following process under the three control methods at this stage. It is evident that V1 performs the worst across all metrics and is not recommended. In contrast, V2 shows the best performance in terms of system safety, although its efficiency is lower than that of inventory control. Overall, during the 80%–30 % output power stage, inventory control remains the preferred method as it ensures the highest efficiency while maintaining system safety and stable operation. However, for stricter system safety requirements, V2 is a suitable choice.

4.2.3. Third stage:30%–0%

According to the previous section, the maximum regulation capacity of the inventory control is reached at 30 % output power. Therefore, only two valve control methods are analyzed in the third stage. Before 1500 s, the system is in a steady-state condition at 30 % output power using inventory control. The variations in each parameter are shown in Fig. 21. Both V1 and V2 methods achieve fast load following, but V2 shows fluctuations during the transient process.

Initially, the compressor mass flow rate is 7.18 kg/s, while the one through the core and turbine is 5.63 kg/s. In V2 method, after the turbine bypass valve opens, the mass flow rate through the core and compressor rises and stabilizes to the same value, while the one at the turbine inlet decreases. Additionally, as shown in Fig. 21h, V1 maintains a surge margin of 20 % by anti-surge protection, while V2 causes a sharp surge margin increase, leading to return valve closure. This explains why V2 method's characteristics are like the previous, while the V1 method presents different characteristics. Additionally, the rapid change in mass flow rate under V2 method may have triggered the fluctuations. Regarding temperature, special attention must be paid to the V1 method. In this process, the maximum core outlet temperature rises to 1358K, 13.17 % higher than the design value, posing a temperature limit risk. Meanwhile, the precooler inlet temperature rises to 628.46K, a 20.66 % rise. Under the V2 method, the trend of the temperature change remains the same, and the compressor inlet temperature returns to 330K due to the return valve closure.

Values of the key parameters at 0 % output power under both methods are shown in Table 7, with a comparative summary of the characteristics in Fig. 22. The system efficiencies are comparable between the V1 and V2 methods, but V2 method outperforms V1 method in all other aspects. However, during the regulation from 100 % to 30 % using inventory control and from 30 % to 0 % using V2 method, slight fluctuations occur due to the switching of the return valve state, though this does not affect the system output power.

4.3. Load-following performance assuming interconnection with renewable energy microgrids

Due to safety concerns, regulators limit the power variation rate, restricting nuclear reactors' load-following capability. Requirements in the European Utilities Requirements (EUR) document [50] is shown in Table 8. Regarding the technology evolving and more pressing need for nuclear reactors to quickly adjust their output power in the future, regulating limitations may become more accommodating. Therefore, in this study, the system is required to follow larger step changes and sharper ramp changes, as shown in Table 8.

The analysis in Section 4.2 shows that the compressor bypass valve control is not suitable for the load-following process, whereas the turbine bypass valve control and the inventory control at the initial stage show better performance. Therefore, three regulation combinations are applied to step and ramp change to verify the load-following capability of the system in this section, listed in Table 9.

4.3.1. Load-following performance under assumed ramp changes

The system dynamic responses for the three regulation combinations

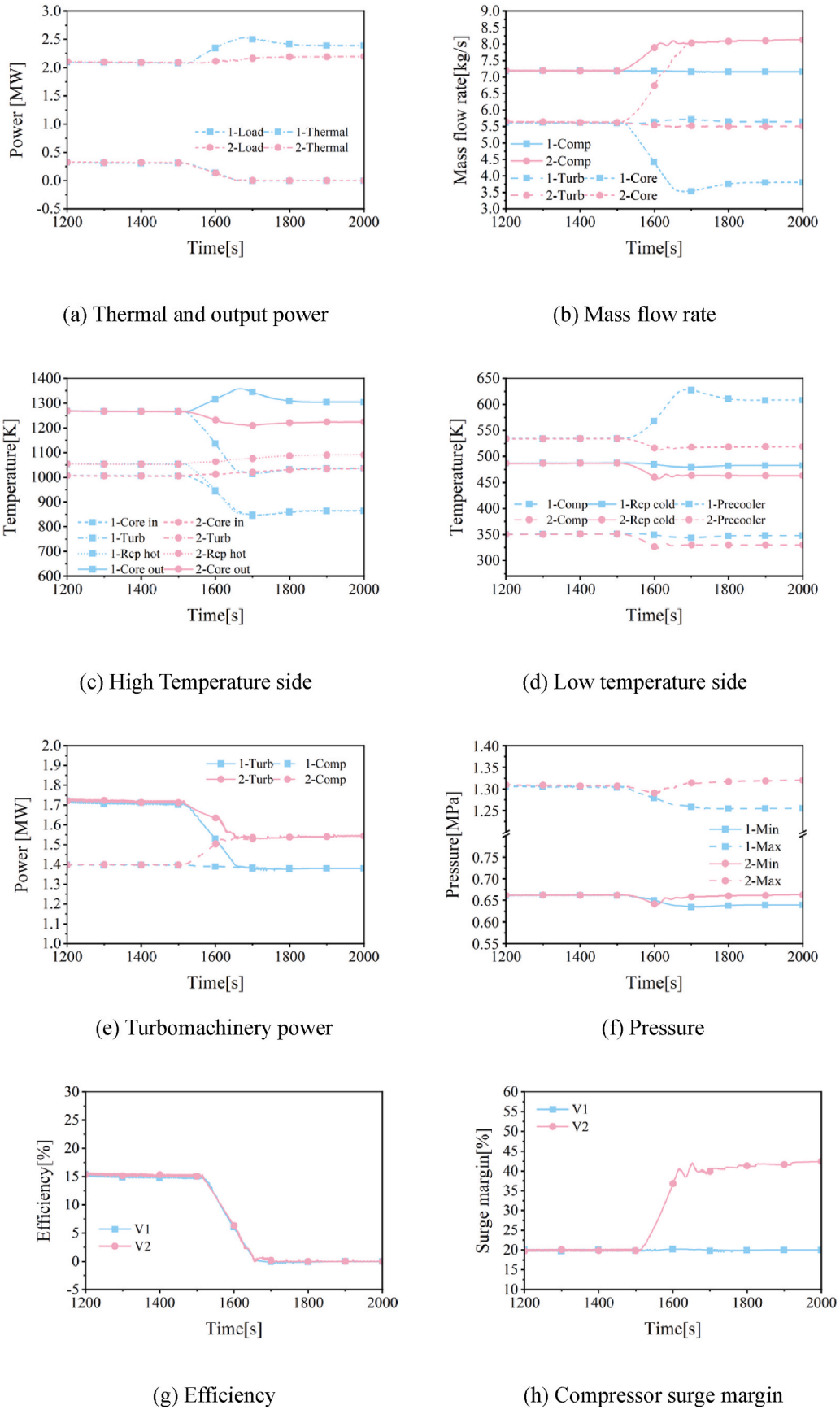


Fig. 21. Close-loop simulation for two control methods with output power from 30 % to 0 %.

Table 7
Parameters at 0 % output power of two control methods.

	30 % c condition	V1(compressor bypass valve)	V2(turbine bypass valve)
Max Temperature [K]	1266.01	1291.68	1224.35
Min Temperature [K]	349.92	347.84	330.00
Max Pressure [MPa]	1.31	1.26	1.32
Min Pressure [MPa]	0.66	0.64	0.66
Mass flow rate [kg/s]	7.18	5.64	8.13
Surge margin[%]	19.84 %	20.00 %	42.37 %
Load power[MW]	30 %	0 %	0 %
System efficiency [%]	15.33 %	0.00 %	0.00 %

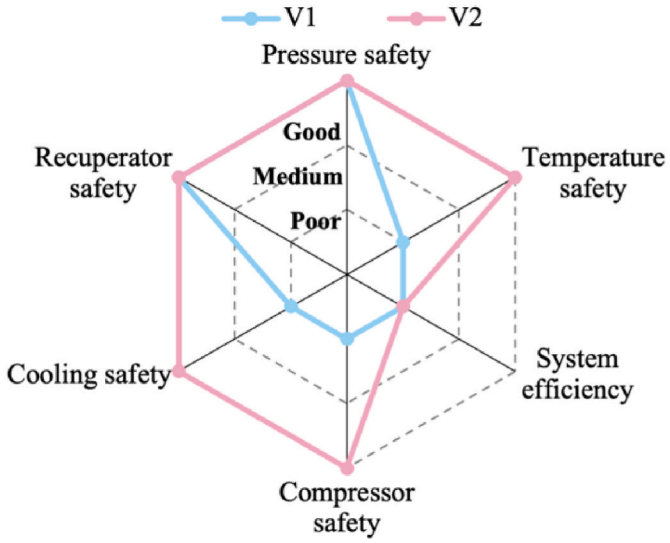


Fig. 22. Performance comparison of two control methods in the process with output power from 30 % to 0 %.

Table 8
Comparison for load-following characteristics.

	EUR Description	In this study
Ramp changes	40 % per hour, equally 20 % in 1800 s	20 % in 100 s
Step changes	20 % in 10 min, equally 20 % in 600 s	20 % in 100 s
Range	100 %–20 % to 100 % in 24 h	100 %–0 % to 100 % in 1200 s

Table 9
Composition of different load-following regulation combinations.

Stage	Single method	Combination one (C1)	Combination two (C2)
100%–80 % output power	V2	Inventory control	Inventory control
80%–30 % output power	V2	V2	Inventory control
30%–0% output power	V2	V2	V2

mentioned above are presented for a load decrease from 100 % to 0 % at an aggressive setpoint ramp rate of 20 % per 100 s (approximately 12 % per minute), with a subsequent increase at the same ramp rate. As seen in Fig. 23a, all three strategies track the load well with no offset or fluctuation, and the changes in system efficiency are depicted in Fig. 23b. The efficiency under strategies involving inventory control (C1, C2) is relatively higher compared to that of the turbine bypass valve (V2) control, consistent with previous observations. Moreover, none of the strategies face the risk of over-temperature or compressor surge, as illustrated in Fig. 23d and e.

4.3.2. Load-following performance under assumed step changes

Different regulation combinations were applied to follow a 20 % step change in output power. The simulation results are shown in Fig. 24. As shown in Fig. 24a, for a step change, all the methods can track the target curve well within a 10-s offset. Meanwhile, the changes in key system parameters are more pronounced, compared to ramp changes, especially for the C1 and C2 regulation strategies. None of them exceed the system safety limits during the transient process. Overall, the analysis shows that all three methods achieve excellent load-following capabilities in both ramp and step changes. Moreover, compared to the two combination strategies, the control strategy using only the turbine bypass valve control (V2) method has less operational complexity, with smoother changes in critical system parameters but relatively lower system efficiency.

5. Conclusion

This paper presents a dynamic model of a HeXe cooled reactor coupled with CBC. Open-loop and closed-loop simulations for different control methods were carried out to evaluate their characteristics and deficiencies within system limitations. Based on these, regulation combinations were tested for step and ramp changes in output power to assess their load-following ability. The main conclusions are as follows.

- The compressor bypass valve (V1) control poses a risk of over-temperature at the core outlet and insufficient precooler cooling capacity due to high precooler inlet temperature, especially in the low power range. It also performs the worst in efficiency.
- The greatest advantage of optimized inventory control is higher efficiency, especially before triggering anti-surge control. However, it must operate in synergy with anti-surge control and faces the risk of increased core outlet temperature.
- The turbine bypass valve (V2) control performs best in all other aspects except efficiency. It is suitable to use alone under strict safety requirements or in combination with inventory control.
- The system can achieve a stable and fluctuation-free ramp change in load at 20 % per 100 s. Additionally, a 20 % step change delay is within a 10-s offset. It can preliminary proof that when integrated with renewable energy, the proposed system can effectively manage the variability and demand fluctuations by load regulation combinations.

This study focuses on the load-following capability of the HeXe cooled SMRs, analyzing the principles and characteristics of different methods, and lays the foundation for integrating SMRs into microgrids. Future work would put the system into microgrids, considering the thermodynamic performance, economic performance, and power demand matching under real operational conditions.

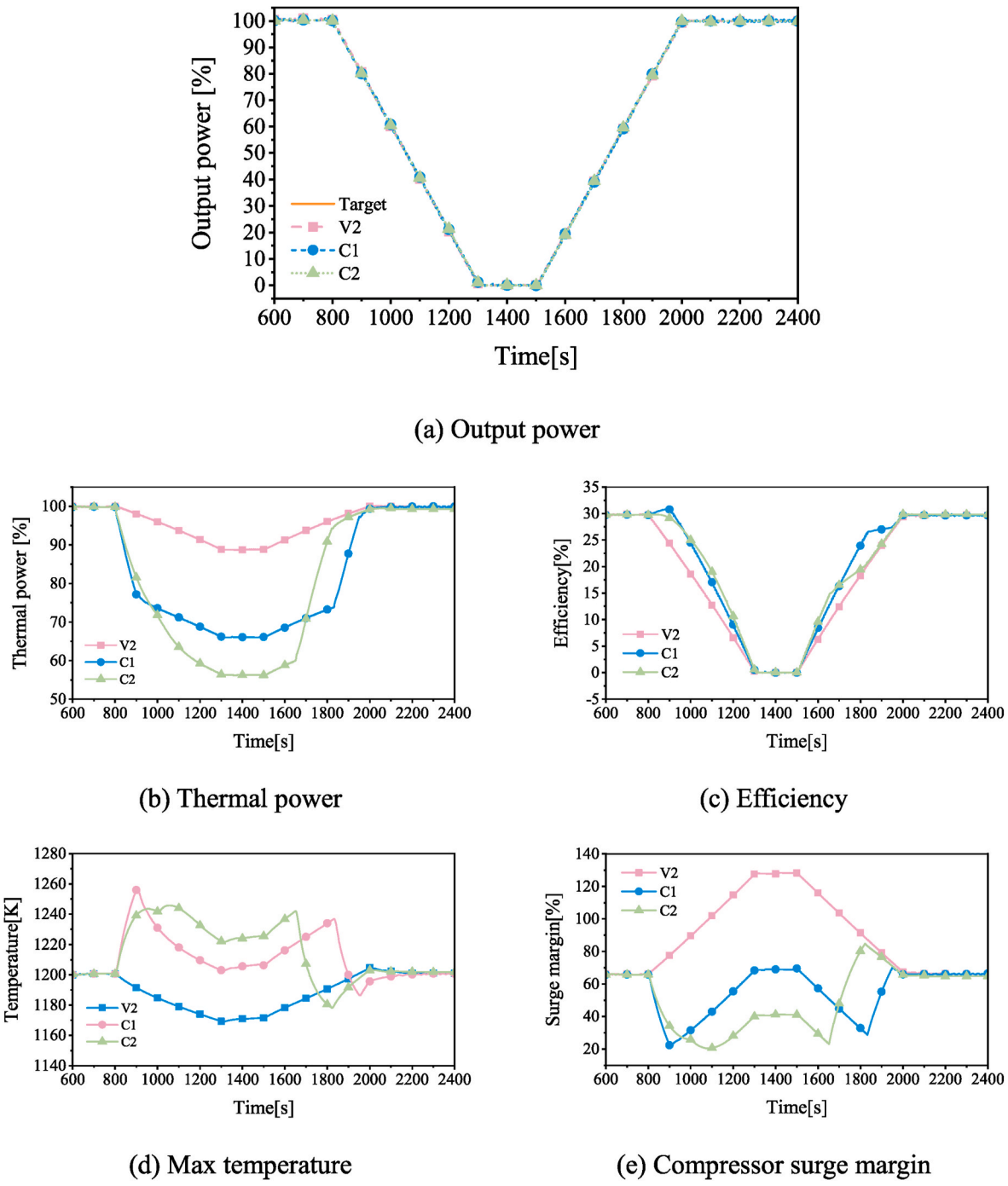


Fig. 23. Load-following performance in ramp change at 20 %/100s in load.

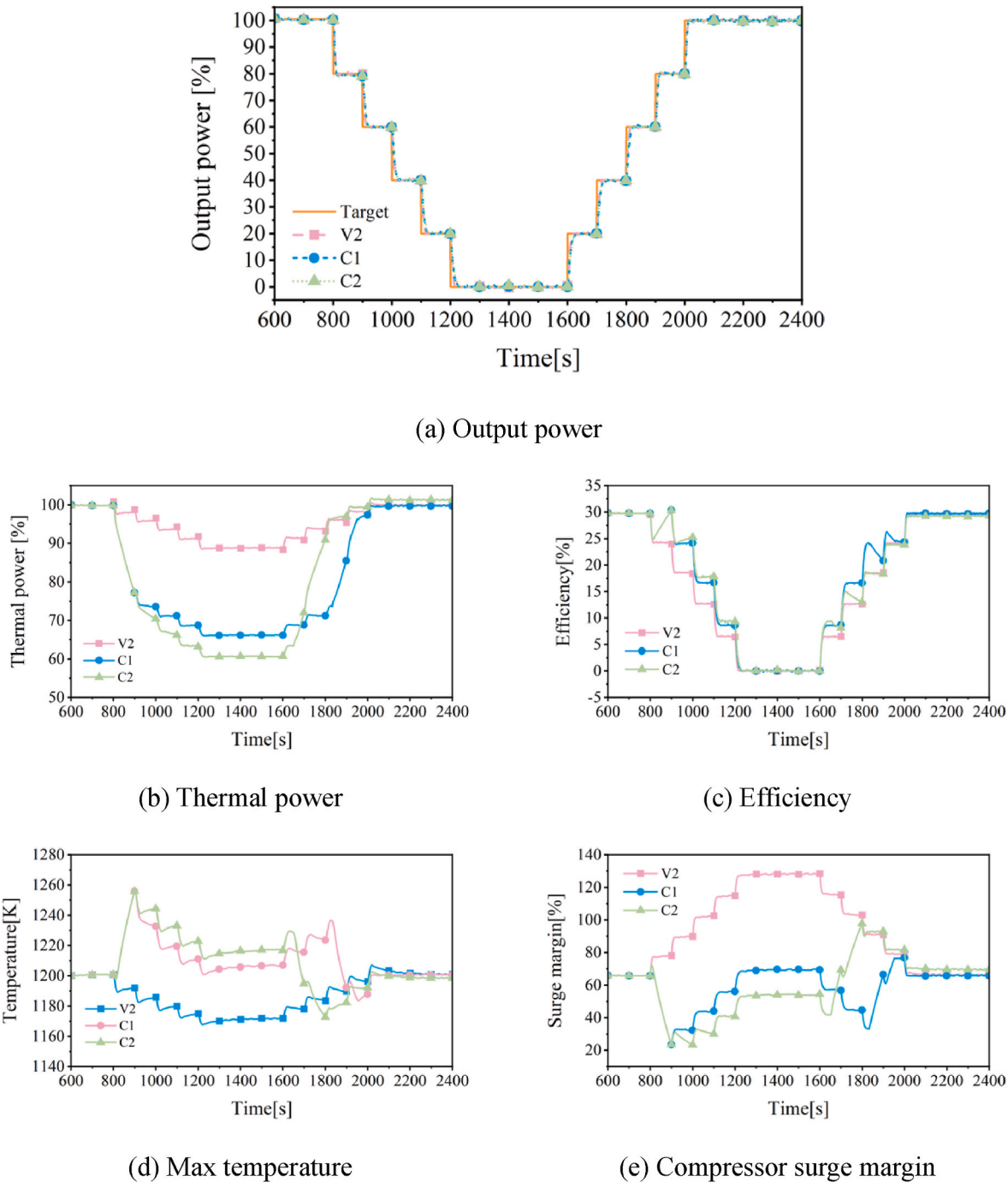


Fig. 24. Load-following performance in 20 % step change in load.

CRediT authorship contribution statement

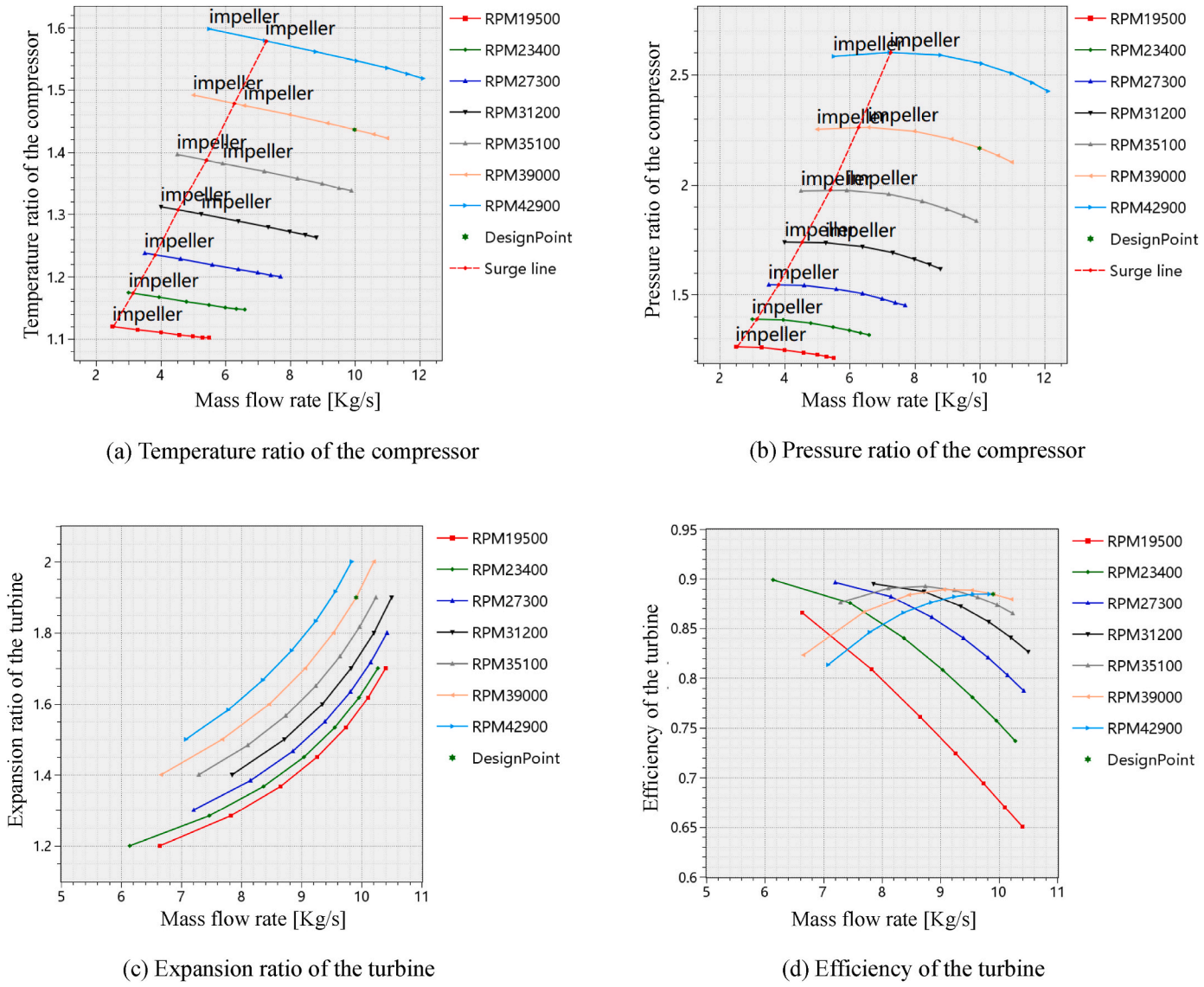
Xinyu Li: Writing – original draft, Software, Methodology, Investigation. **Chaoran Guan:** Validation, Software, Investigation, Formal analysis. **Xiang Chai:** Writing – review & editing, Project administration, Methodology, Funding acquisition. **Xiaojing Liu:** Writing – review & editing, Project administration, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix

Characteristic curves of the turbomachinery are shown in the following Figure A-1.
The PID parameters and variables of control methods used in this study are listed in Table A-1.



(a) Temperature ratio of the compressor

(b) Pressure ratio of the compressor

(c) Expansion ratio of the turbine

(d) Efficiency of the turbine

Fig. A-1. Characteristic curves of the turbomachinery.

Table A-1
The PID parameters and variables of control methods

	PID parameters			Variables	
	K_p	K_i	K_d	Input	Output
Compressor inlet temperature (CIT) control	−0.5	0	−1.2	CIT	Increase in the mass flow rate through the precooler cold side
Anti-surge control	0.1	0	0	Surge margin	Increase in compressor return valve opening
Compressor bypass valve (V1) control	$-1.5\ e^{-7}$	0	0	Output power	Increase in V1 opening
Turbine bypass valve (V2) control	$-1.0\ e^{-7}$	$-1.0\ e^{-6}$	0	Output power	Increase in V2 opening

Data availability

No data was used for the research described in the article.

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